رواد مصر الرقمية

وزارة الاتصـــــالات
وتكنولوجيا المعلومات

Excel 2

Excel

رواد مصر الرقمية

Formulas

- Microsoft Excel is a popular tool for managing data and performing data analysis. It is used for generating analytical reports, business insights, and storing operational records. To perform simple calculations or analyses on data, we need Excel formulas.
- Even simple Excel formulas allow us to manipulate string, number, and date data fields. Furthermore, you can use if-else statements, find and replace, mathematics and trigonometry, finance, logical, and engineering formulas.
- Unlike programming languages, you will be writing the formula name and arguments—nothing complex. You can also use Excel-assisted user interface to add formulas.

SUM Function:

- Adds up all the numbers in a range of cells.
- **Syntax:** = SUM(number1, [number2], ...)
- **Parameters**:
  - number1: The first number or range of numbers to add.
  - [number2]: (Optional) Additional numbers or ranges to add.

Example: =SUM(A1:A10, B1:B10)

adds all values from both ranges A1and B1

SUM Example:

=SUM (number1 , [number2] , ...)

=SUM(C:C)

count Band	Profit	Date
ne	$ 16,185.00	1/1/2014
ne	$ 13,210.00	1/1/2014
ne	$ 10,890.00	6/1/2014
ne	$ 4,440.00	6/1/2014
ne	$ 12,350.00	6/1/2014
ne	$ 136,170.00	12/1/2014
ne	$ 4,605.00	3/1/2014
ne	$ 22,662.00	6/1/2014
ne	$ 18,990.00	6/1/2014
ne	$ 13,905.00	6/1/2014
ne	$ 12,350.00	6/1/2014
ne	$ 13,327.50	7/1/2014
ne	$ 47,900.00	8/1/2014
ne	$ 4,292.00	9/1/2014
ne	$ 1,725.00	10/1/2013
ne	$ 3,075.00	12/1/2014
ne	$ 2,920.00	2/1/2014
ne	$ 4,870.00	2/1/2014
ne	$ 22,662.00	6/1/2014
ne	$ 90,540.00	6/1/2014
ne	$ 3,303.00	7/1/2014
ne	$ 1,766.00	8/1/2014

WE WANT THE TOTAL PROFITS

=SUM(C:C)

AVERAGE Function:

- Calculates the average in a range of cells.

- **Syntax:** =AVERAGE(number1, [number2], ...)

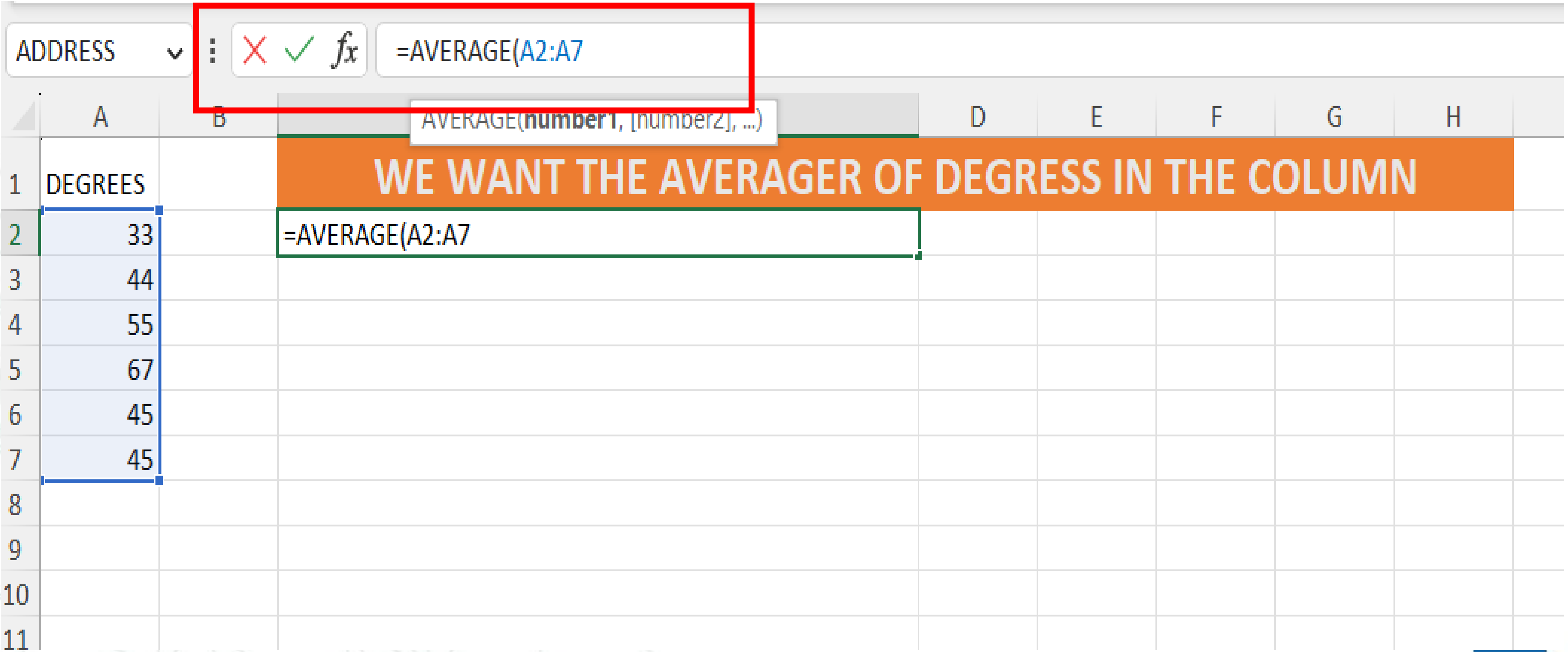
- **Parameters:**
  - number1: The first number or range of numbers to calculate the average.
  - [number2]: (Optional) Additional numbers or ranges to include in the average calculation.

Example: =AVERAGE(A1:A10, B1:B10)

calculates the average of all values in ranges A1and B1

AVERAGE

=AVERAGE (number1 , [number2] , ...)



MAX and MIN Function:

- **Syntax:**
  - =MAX(number1, [number2], ...)
  - =MIN(number1, [number2], ...)
- **Parameters:**
  - number1: The first number or range of numbers to evaluate.
  - [number2]: (Optional) Additional numbers or ranges to evaluate.

Examples:

=MAX(A1:A10, B1:B10) finds the maximum value from both ranges A1and B1.

=MIN(A1:A10, B1:B10) finds the minimum value from both ranges A1and B1.

MAX and MIN Example:

=MAX (number1 , [number2] , ...)

=MIN (number1 , [number2] , ...)

C7

	A	B	C	D
1	DEGREES		WE WANT THE MAXIMUM&MINIMUM DEGREE IN THE COLUMN	
2	33			
3	44		MIN	33
4	55		MAX	67
5	67			
6	45			
7	45			
8				
9				
10				
11				
12				
13				

LARGE and SMALL Functions:

- **Syntax:**
  - =LARGE(array, k)
  - =SMALL(array, k)
- **Parameters:**
  - array: The range or array of numbers to evaluate.
  - k: The position (1st, 2nd, 3rd, etc.) of the value to return from the smallest values.
- **Example:**
  - =LARGE(A1:A10, 2) returns the second-largest value in the range A1.
  - =SMALL(A1:A10, 2) returns the second-smallest value in the range A1.

LARGE and SMALL Example:

=SMALL (array , k)

=LARGE (array , k)

C2 | =LARGE(A2:A11,3)

	A	B	C
1	DEGREE		
2	12		70
3	33		33
4	44		
5	55		
6	25		
7	64		
8	70		
9	67		
10	80		
11	99		

COUNT Function:

- **Description:**
  - Counts the number of cells that contain **numerical values** within a range.
- **Syntax:** =COUNT(value1, [value2], ...)
- **Parameters:**
  - value1: The first range or value to count.
  - [value2]: (Optional) Additional ranges or values to include in the count.

Example: =COUNT(A1:A10)

counts the number of cells with numerical values in the range A1.

COUNTA Function:

- **Description:**
  - Counts the number of cells that **are not empty within a range**, including cells with numbers, text, or any other type of data.
- **Syntax:** =COUNTA(value1, [value2], ...)
- **Parameters:**
  - value1: The first range or value to count.
  - [value2]: (Optional) Additional ranges or values to include in the count.
- **Example:** =COUNTA(A1:A10)

counts the number of non-empty cells in the range A1

COUNTBLANK Function:

- **Description:**
  - Counts the number of empty cells within a range.
- **Syntax:** =COUNTBLANK(range)
- **Parameters:**
  - range: The range within which to count the empty cells.
- **Example:** =COUNTBLANK(A1:A10)

  counts the number of empty cells in the range A1

COUNT, COUNTA and COUNTBLANK:

=COUNT (number1 , [number2] , ...)

=COUNTA (number1 , [number2] , ...)

=COUNTBLANK (number1 , [number2] , ...)

	A	B	C	D	E	F	G	H	I
1	NAMES	AGE	COUNT WE USE TO COUNT THE CELL IN NUMERICAL						
2	ALI	12	9						
3	SAID	33	COUNTA WE USE IT TO COUNT NUMERICAL OR NON NUMERICAL WITHOUT BLANCK VALUES						
4		44	7						
5	MONIR	55	COUNT BLANCK TO COUNT THE NULL VALUES						
6	TAMER	63	2						
7	GAMAL	34							
8	SABER	32							
9		42							
10	SOLIMAN	22							

COUNTIF Function:

- **Description**: Counts the number of cells within a range that **meet a single condition**.
- **Syntax**: =COUNTIF(range, criteria)
- **Parameters**:
  - **range**: The range of cells to apply the criteria to.
  - **criteria**: The condition that must be met. It can be a number, text, or expression.

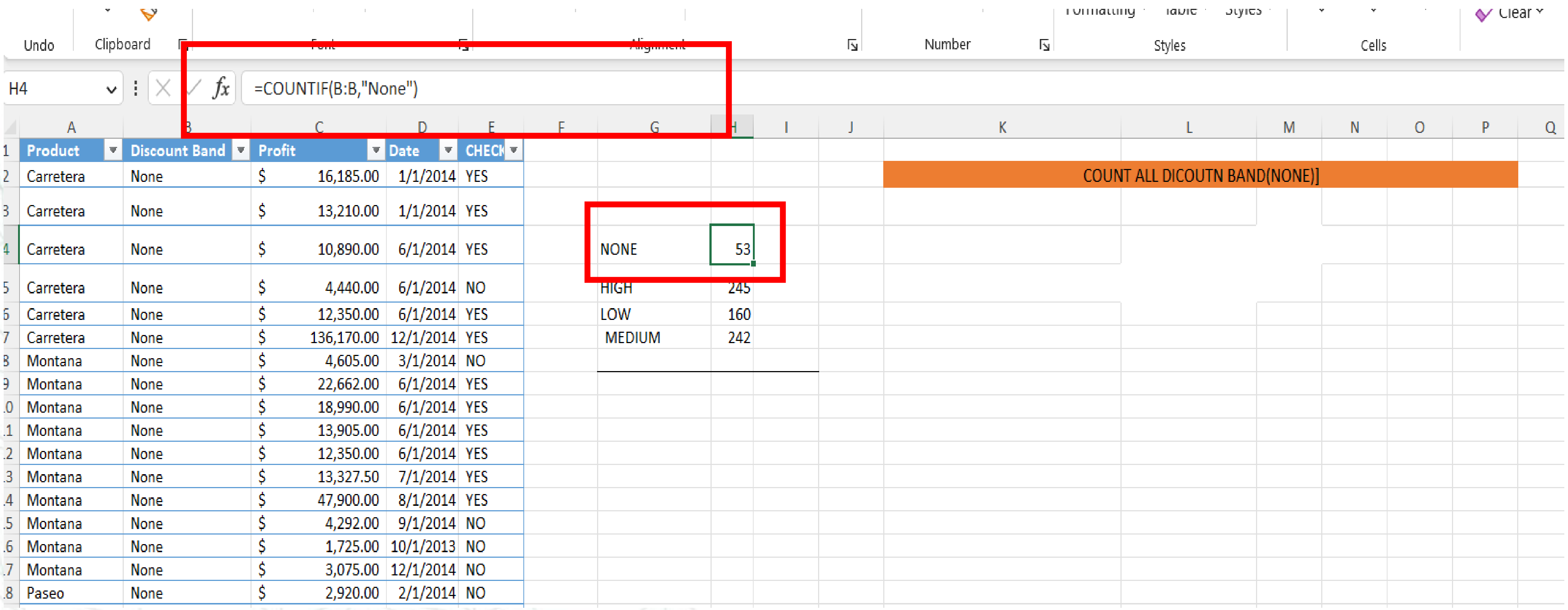
Example:

=COUNTIF(A1:A10, "apple")

Counts the number of cells in the range A1 that contain the value "apple".

COUNTIF Function:

=COUNTIF (range , criteria)



Product	Discount Band	Profit	Date	CHECK		G	H
Carretera	None	$ 16,185.00	1/1/2014	YES			
Carretera	None	$ 13,210.00	1/1/2014	YES			
Carretera	None	$ 10,890.00	6/1/2014	YES		NONE	53
Carretera	None	$ 4,440.00	6/1/2014	NO		HIGH	245
Carretera	None	$ 12,350.00	6/1/2014	YES		LOW	160
Carretera	None	$ 136,170.00	12/1/2014	YES		MEDIUM	242
Montana	None	$ 4,605.00	3/1/2014	NO			
Montana	None	$ 22,662.00	6/1/2014	YES			
Montana	None	$ 18,990.00	6/1/2014	YES			
Montana	None	$ 13,905.00	6/1/2014	YES			
Montana	None	$ 12,350.00	6/1/2014	YES			
Montana	None	$ 13,327.50	7/1/2014	YES			
Montana	None	$ 47,900.00	8/1/2014	YES			
Montana	None	$ 4,292.00	9/1/2014	NO			
Montana	None	$ 1,725.00	10/1/2013	NO			
Montana	None	$ 3,075.00	12/1/2014	NO			
Paseo	None	$ 2,920.00	2/1/2014	NO			

COUNTIFS Function:

- **Description**: Counts the number of cells that meet multiple criteria **across different ranges.**
- **Syntax**: =COUNTIFS(criteria_range1, criteria1, [criteria_range2, criteria2], ...)
- **Parameters**:
  - **criteria_range1**: The range of cells to evaluate for the first criterion.
  - **criteria1**: The condition that the cells in criteria_range1 must meet.
  - **criteria_range2, criteria2, ...**: Additional ranges and criteria.

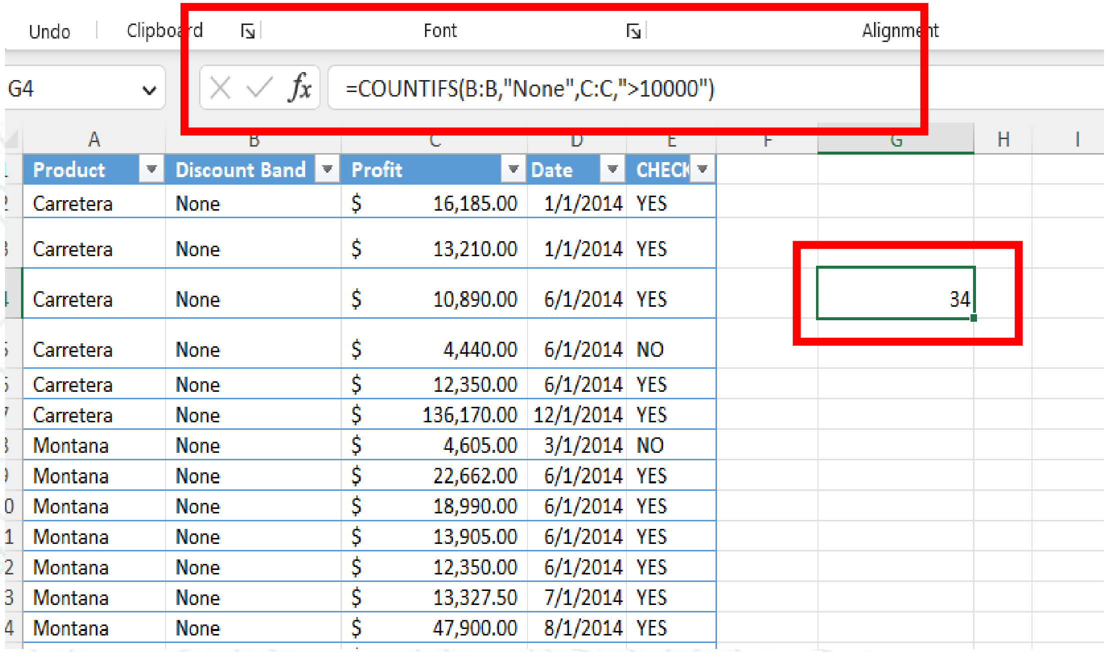
Example:

=COUNTIFS(A1:A10, "apple", B1:B10, ">5")

Counts the number of rows where the value in column A is "apple" and the value in column B is greater than 5.

COUNTIFS Examples:

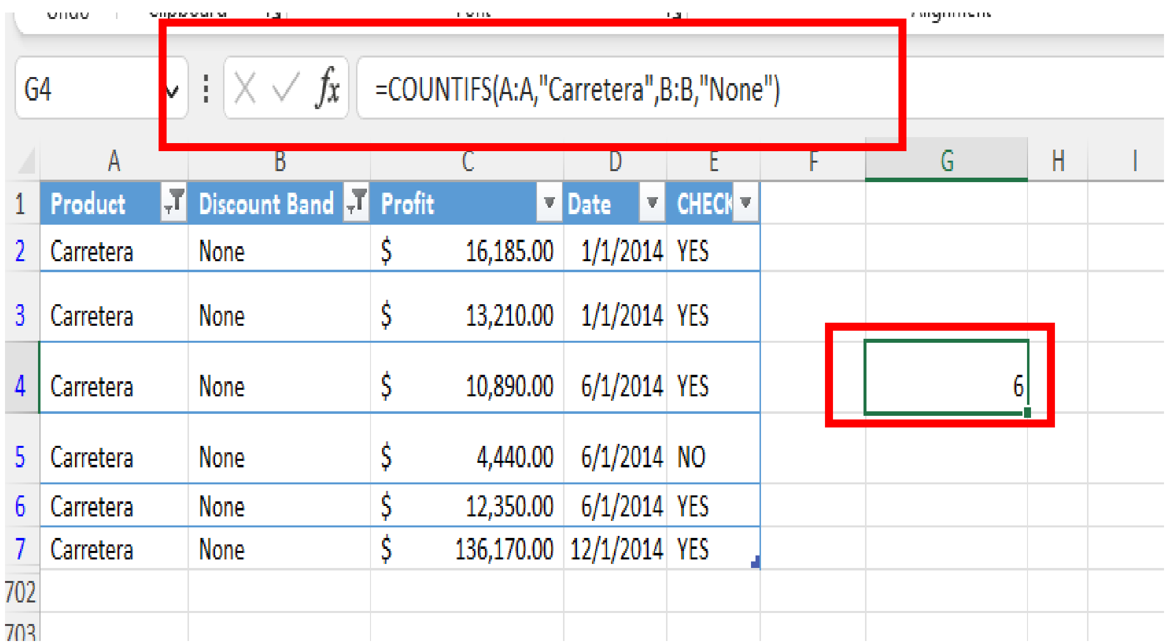
=COUNTIFS (criteria_range1 , criteria1 , [criteria_range2] , [criteria2] , ...)



G4 =COUNTIFS(B:B,"None",C:C,">10000")

Product	Discount Band	Profit	Date	CHECK
Carretera	None	$ 16,185.00	1/1/2014	YES
Carretera	None	$ 13,210.00	1/1/2014	YES
Carretera	None	$ 10,890.00	6/1/2014	YES
Carretera	None	$ 4,440.00	6/1/2014	NO
Carretera	None	$ 12,350.00	6/1/2014	YES
Carretera	None	$ 136,170.00	12/1/2014	YES
Montana	None	$ 4,605.00	3/1/2014	NO
Montana	None	$ 22,662.00	6/1/2014	YES
Montana	None	$ 18,990.00	6/1/2014	YES
Montana	None	$ 13,905.00	6/1/2014	YES
Montana	None	$ 12,350.00	6/1/2014	YES
Montana	None	$ 13,327.50	7/1/2014	YES
Montana	None	$ 47,900.00	8/1/2014	YES

G4: 34



G4 =COUNTIFS(A:A,"Carretera",B:B,"None")

Row	Product	Discount Band	Profit	Date	CHECK
2	Carretera	None	$ 16,185.00	1/1/2014	YES
3	Carretera	None	$ 13,210.00	1/1/2014	YES
4	Carretera	None	$ 10,890.00	6/1/2014	YES
5	Carretera	None	$ 4,440.00	6/1/2014	NO
6	Carretera	None	$ 12,350.00	6/1/2014	YES
7	Carretera	None	$ 136,170.00	12/1/2014	YES
702					
703					

G4: 6

SUMIF Function:

- **Description**: Adds the cells specified by a **given condition or criteria**.
- **Syntax**: =SUMIF(range, criteria, [sum_range])
- **Parameters**:
  - **range**: The range of cells to evaluate.
  - **criteria**: The condition that determines which cells to add.
  - **sum_range**: The actual cells to sum. If omitted, the cells in the range are summed.

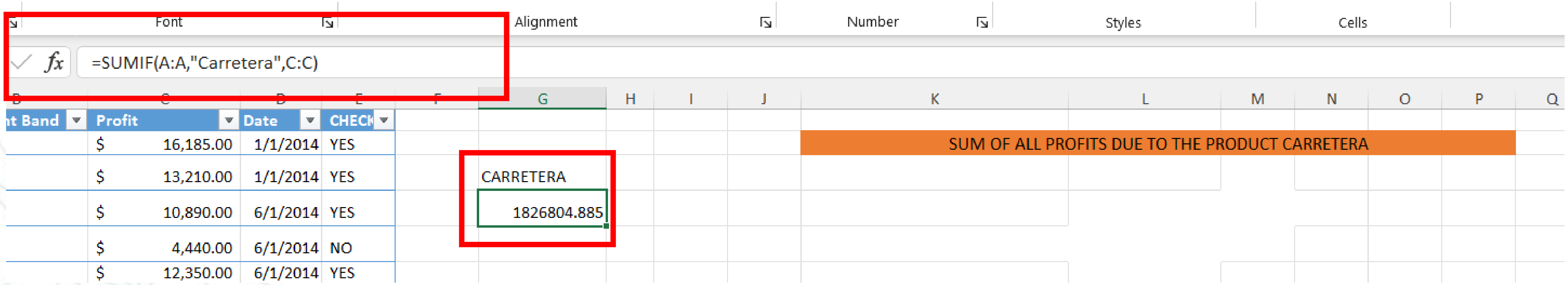
Example:

=SUMIF(A1:A10, "apple", C1:C10)

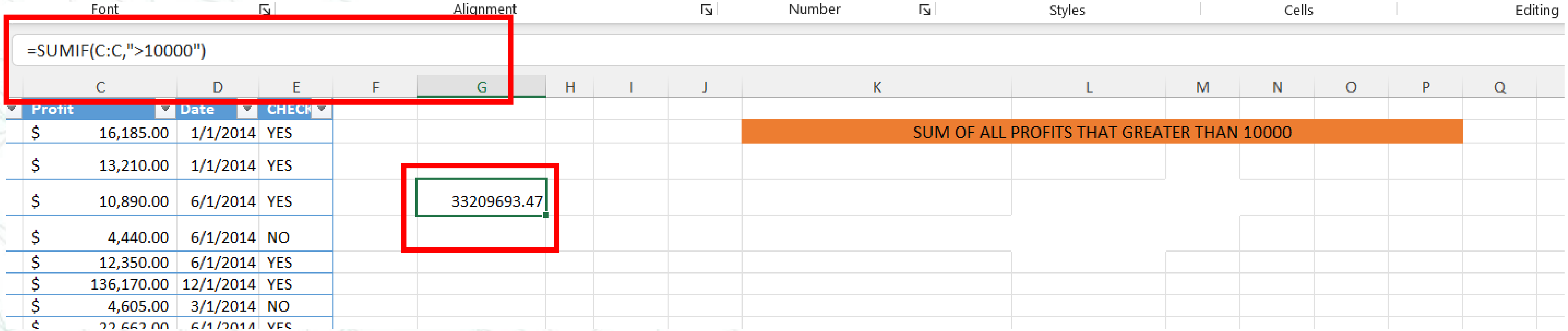
Sums the values in range C1 where corresponding cells in A1 are "apple".

SUMIF Examples:

=SUMIF (range , criteria , [sum_range])



Profit	Date	CHECK
$ 16,185.00	1/1/2014	YES
$ 13,210.00	1/1/2014	YES
$ 10,890.00	6/1/2014	YES
$ 4,440.00	6/1/2014	NO
$ 12,350.00	6/1/2014	YES



Profit	Date	CHECK
$ 16,185.00	1/1/2014	YES
$ 13,210.00	1/1/2014	YES
$ 10,890.00	6/1/2014	YES
$ 4,440.00	6/1/2014	NO
$ 12,350.00	6/1/2014	YES
$ 136,170.00	12/1/2014	YES
$ 4,605.00	3/1/2014	NO

SUMIFS Function:

- **Description**: Adds the cells specified by **multiple criteria**.
- **Syntax**: =SUMIFS(sum_range, criteria_range1, criteria1, [criteria_range2, criteria2], ...)
- **Parameters**:
  - **sum_range**: The cells to sum.
  - **criteria_range1**: The range to apply the first criteria to.
  - **criteria1**: The first condition to be met.
  - **criteria_range2, criteria2, ...**: Additional ranges and criteria.

Example: =SUMIFS(D1:D10, A1:A10, "apple", B1:B10, ">5")

Sums the values in range D1 where corresponding cells in A1 are "apple" and cells in B1 are greater than 5.

SUMIFS Example:

=SUMIFS (sum_range , criteria_range1 , criteria1 , [criteria_range2] , [criteria2] , …..)

=SUMIFS(C:C,A:A,"Carretera",B:B,"None")

and	Profit	Date	CHECK
	$ 16,185.00	1/1/2014	YES
	$ 13,210.00	1/1/2014	YES
	$ 10,890.00	6/1/2014	YES
	$ 4,440.00	6/1/2014	NO
	$ 12,350.00	6/1/2014	YES
	$ 136,170.00	12/1/2014	YES
	$ 4,605.00	3/1/2014	NO
	$ 22,662.00	6/1/2014	YES
	$ 18,990.00	6/1/2014	YES
	$ 13,905.00	6/1/2014	YES
	$ 12,350.00	6/1/2014	YES
	$ 13,327.50	7/1/2014	YES

193245

SUM OF ALL PROFITS WHEN THE PRODUCT IS CERRETERA AND DICOUBT BAND IS NONE

AVERAGEIF Function:

- **Description**: Calculates the average of cells that **meet a single condition**.
- **Syntax**: =AVERAGEIF(range, criteria, [average_range])
- **Parameters**:
  - **range**: The range of cells to apply the criteria to.
  - **criteria**: The condition that must be met.
  - **average_range**: The actual cells to average. If omitted, the cells in the range are averaged.

Example: =AVERAGEIF(A1:A10, "apple", E1:E10)

Averages the values in range E1 where corresponding cells in A1 are "apple".

AVERAGEIF Example:

=AVERAGEIF (range , criteria , [average_range])

Font | Alignment | Number | Styles | Cells

fx =AVERAGEIF(A:A,"Paseo",C:C)

d	Profit	Date	CHECK	F	G	H	I	J	K
	$ 16,185.00	1/1/2014	YES						AVEGARGE OF PROFITS WHEN THE PRODUCT IS PASEO
	$ 13,210.00	1/1/2014	YES		23749.69282				
	$ 10,890.00	6/1/2014	YES						
	$ 4,440.00	6/1/2014	NO						
	$ 12,350.00	6/1/2014	YES						
	$ 136,170.00	12/1/2014	YES						
	$ 4,605.00	3/1/2014	NO						
	$ 22,662.00	6/1/2014	YES						
	$ 18,990.00	6/1/2014	YES						
	$ 13,905.00	6/1/2014	YES						
	$ 12,350.00	6/1/2014	YES						
	$ 13,327.50	7/1/2014	YES						
	$ 47,900.00	8/1/2014	YES						

AVERAGEIFS

- **Description**: Calculates the average of cells **that meet multiple criteria**.
- **Syntax**: =AVERAGEIFS(average_range, criteria_range1, criteria1, [criteria_range2, criteria2], ...)
- **Parameters**:
  - **average_range**: The cells to average.
  - **criteria_range1**: The range to apply the first criteria to.
  - **criteria1**: The first condition to be met.
  - **criteria_range2, criteria2, ...**: Additional ranges and criteria.

Example: =AVERAGEIFS(F1:F10, A1:A10, "apple", B1:B10, ">5")

Averages the values in range F1 where corresponding cells in A1 are "apple" and cells in B1 are greater than 5.

AVERAGEIFS

```
=AVERAGEIFS(average_range, criteria_range1, criteria1, ...)
```

=AVERAGEIFS(C:C,A:A,"Paseo",B:B,"None")

Profit	Date	CHECK
$ 16,185.00	1/1/2014	YES
$ 13,210.00	1/1/2014	YES
$ 10,890.00	6/1/2014	YES
$ 4,440.00	6/1/2014	NO
$ 12,350.00	6/1/2014	YES
$ 136,170.00	12/1/2014	YES

34094

AVEGARGE OF PROFITS WHEN THE PRODUCT IS PASEO AND DISCOUJNTBAND IS HIGH

MODE Function:

Description: Returns the most frequently occurring value in a data set. If there are multiple values with the same highest frequency, MODE returns the smallest of those values.

Syntax: =MODE(number1, [number2], ...)

Parameters:

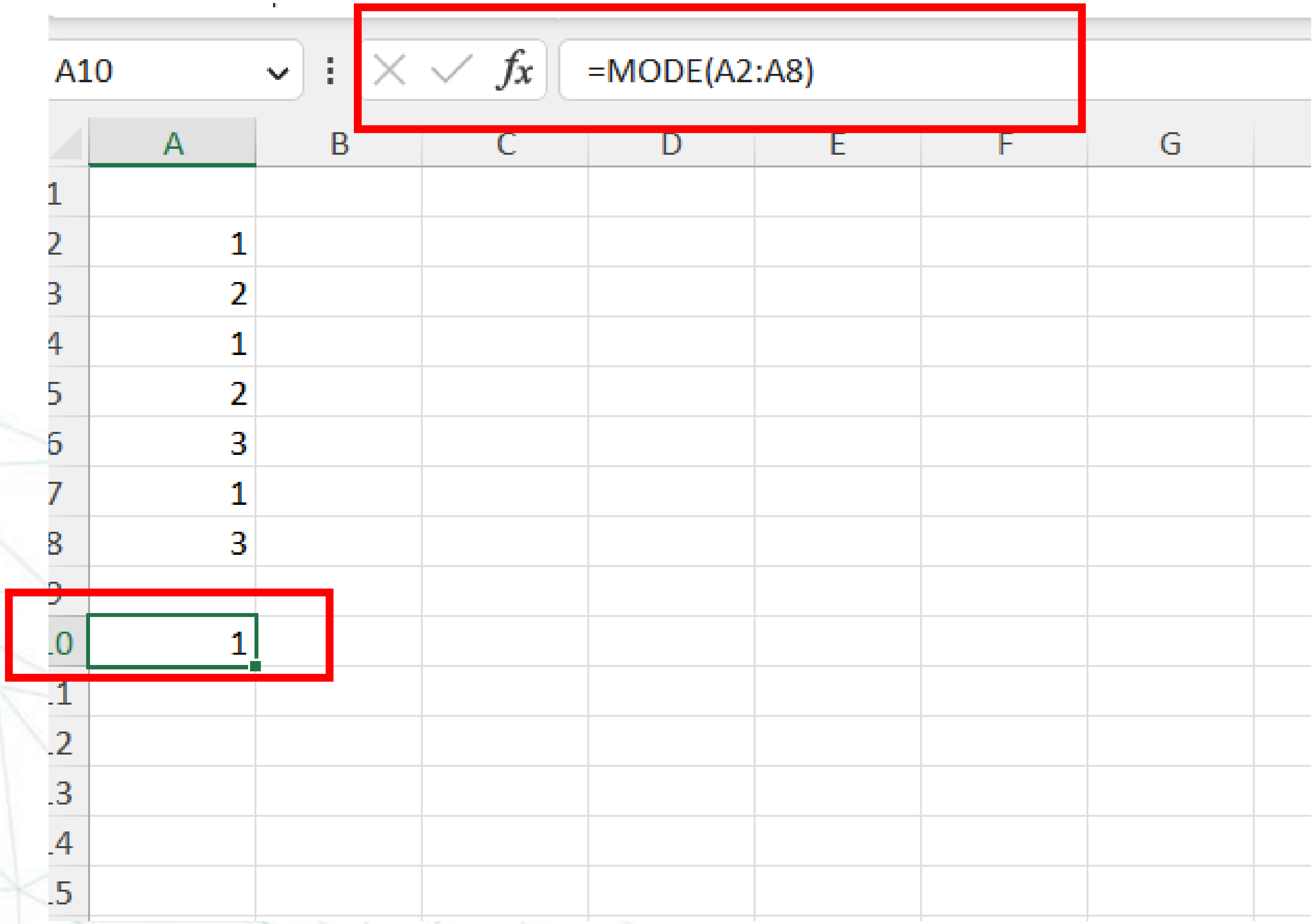
- **number1, number2, ...**: The numbers or ranges of cells from which to determine the most frequently occurring value. You can specify up to 255 numbers or ranges.

Example: If you have a data set in cells A1 and you want to find the most frequently occurring value:

=MODE(A1:A10)

Returns the value that appears most frequently in the range A1

MODE



VLOOKUP Function:

- **Description**: Searches for a value in the first column of a table and returns a value in the same row from another column.
- **Syntax**: =VLOOKUP(lookup_value, table_array, col_index_num, [range_lookup])
- **Parameters**:
  - **lookup_value**: The value to search for.
  - **table_array**: The range of cells that contains the data.
  - **col_index_num**: The column number in the table from which to retrieve the value.
  - **[range_lookup]**: TRUE for approximate match or FALSE for exact match.

Example: =VLOOKUP("apple", A1:C10, 2, FALSE)

Finds "apple" in the first column of the range A1 and returns the value from the second column in the same row.

VLOOKUP Example:

E2 =VLOOKUP(E1, A2:B11, 2, FALSE)

	A	B
1	**Animal**	**Speed (mph)**
2	Cheetah	70
3	Antelope	61
4	Lion	50
5	Elk	45
6	Coyote	43
7	Gray fox	42
8	Ostrich	40
9	Greyhound	39.35
10	Whippet	35.5
11	Rabbit	35

	D	E
1	**Animal**	Lion
2	**Speed**	50

=VLOOKUP(E1, A2:B11, 2, FALSE)

Look up this value, in this range, return a match from this column, search for exact match

HLOOKUP Function:

- **Description**: Searches for a value in the top row of a table and returns a value in the same column from another row.
- **Syntax:** =HLOOKUP(lookup_value, table_array, row_index_num, [range_lookup])
- **Parameters**:
  - **lookup_value**: The value to search for.
  - **table_array**: The range of cells that contains the data.
  - **row_index_num**: The row number in the table from which to retrieve the value.
  - **[range_lookup]**: TRUE for approximate match or FALSE for exact match.

Example: =HLOOKUP("Jan", A1:D10, 3, FALSE)

Finds "Jan" in the top row of the range A1 and returns the value from the third row in the same column.

HLOOKUP Example:

C7 =HLOOKUP("Joey",A1:G5,5,FALSE)

	A	B	C	D	E	F	G	H
1	Student's Name	Monica	Rachel	Phoebe	Joey	Ross	Chandler	
2	Maths	76	53	67	57	81	70	
3	Science	86	43	65	47	91	68	
4	English	87	44	72	48	92	75	
5	General Awareness	92	56	98	60	97	101	
6								
7			=HLOOKUP("Joey",A1:G5,5,FALSE)					
8								

1	Student's Name	Monica	Rachel	Phoebe	Joey	Ross	Chandler
2	Maths	76	53	67	57	81	70
3	Science	86	43	65	47	91	68
4	English	87	44	72	48	92	75
5	General Awareness	92	56	98	60	97	101
6							
7			60				
8							

IF Function:

- **Description**: Checks whether a condition is met, returns one value if TRUE and another if FALSE.
- **Syntax**: =IF(logical_test, value_if_true, value_if_false)
- **Parameters**:
  - **logical_test**: The condition to test.
  - **value_if_true**: The value to return if the condition is TRUE.
  - **value_if_false**: The value to return if the condition is FALSE.

Example: =IF(A1 > 10, "Greater", "Lesser")

Checks if the value in cell A1 is greater than 10. Returns "Greater" if TRUE, otherwise "Lesser".

IF Example:

D2 | =IF(AND(B2>50, C2>50), "Pass", "Fail")

	A	B	C	D	E
1	**Name**	**Test 1**	**Test 2**	**Result**	
2	Colin	57	62	Pass	
3	Den	38	63	Fail	
4	Dustin	43	72	Fail	
5	Frank	77	69	Pass	
6	James	76	49	Fail	
7	Karl	66	74	Pass	
8	Mike	93	58	Pass	
9	Scott	61	84	Pass	
10	Yvette	50	50	Fail	

Data Visualization

Assignment

- **Task 1:** Use Excel to analyze a dataset (e.g., sales or inventory data) by applying functions such as SUM, AVERAGE, MIN, MAX, and IF.
- **Task 2:** Combine Excel functions (e.g., SUMIF, COUNTIF) to answer specific business questions from the dataset.
- **Task 3:** Create a report with screenshots showing how you applied each function and the resulting output.

Data visualization

is the graphical representation of information and data. By using visual elements like charts, graphs, and maps, data visualization tools provide an accessible way to see and understand trends, outliers, and patterns in data.

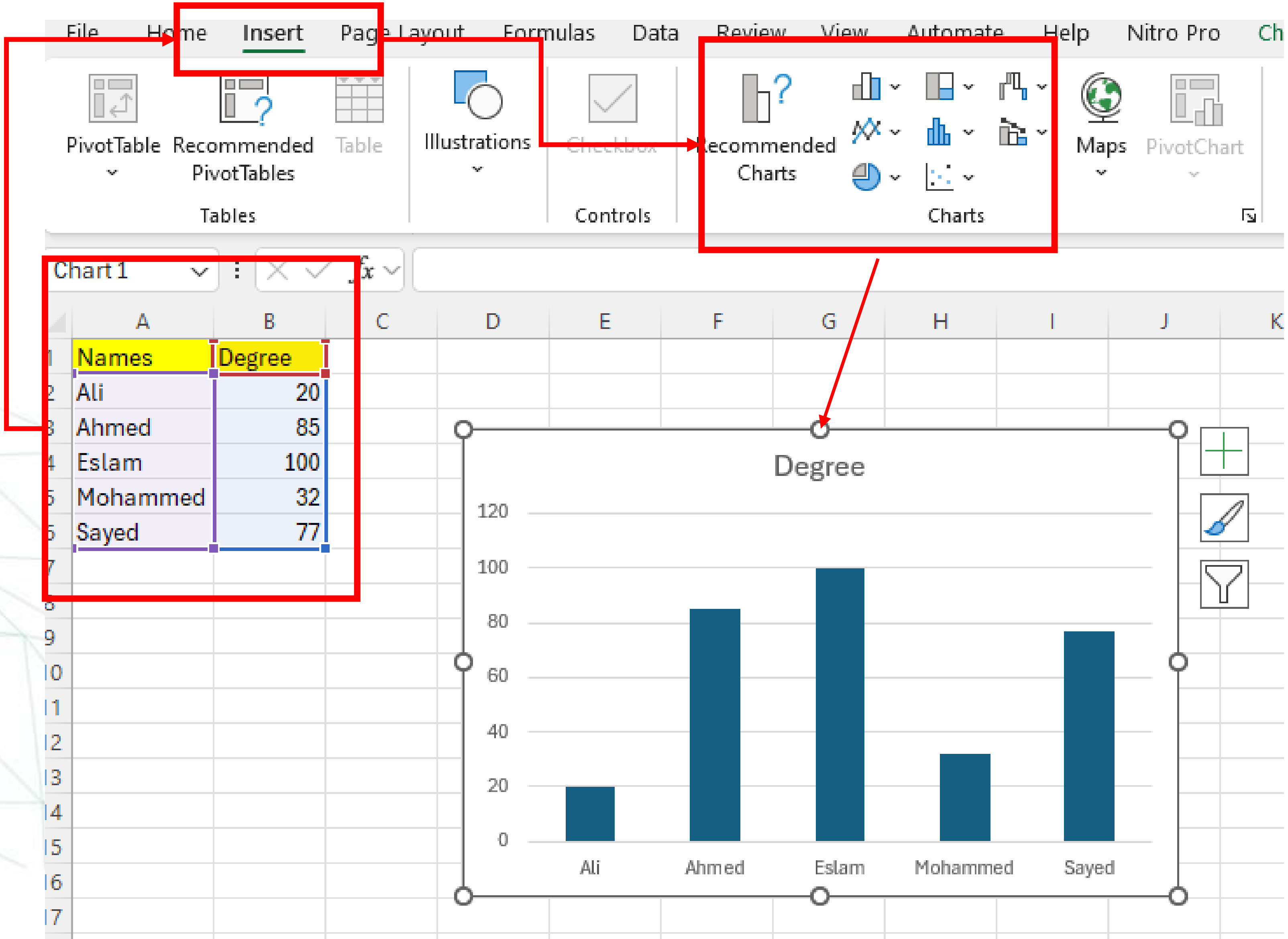
Additionally, it offers an excellent means for employees or business owners to present data to non-technical audiences without causing confusion.

First Step With Data Visualization

- In the world of Big Data, data visualization tools and technologies are essential for analyzing massive amounts of information and making data-driven decisions.
- Let's start with data visualization. How can we create a chart to represent data?

1. **Select the data**.
2. Go to **Insert**.
3. Choose the best chart for the data.

If you have this simple data



Notice what happened when you insert a chart in Excel, two additional tabs appear on the Ribbon:

Format

Design

These tabs give you tools to customize the appearance and layout of your chart.

Additionally, three buttons appear beside the chart:

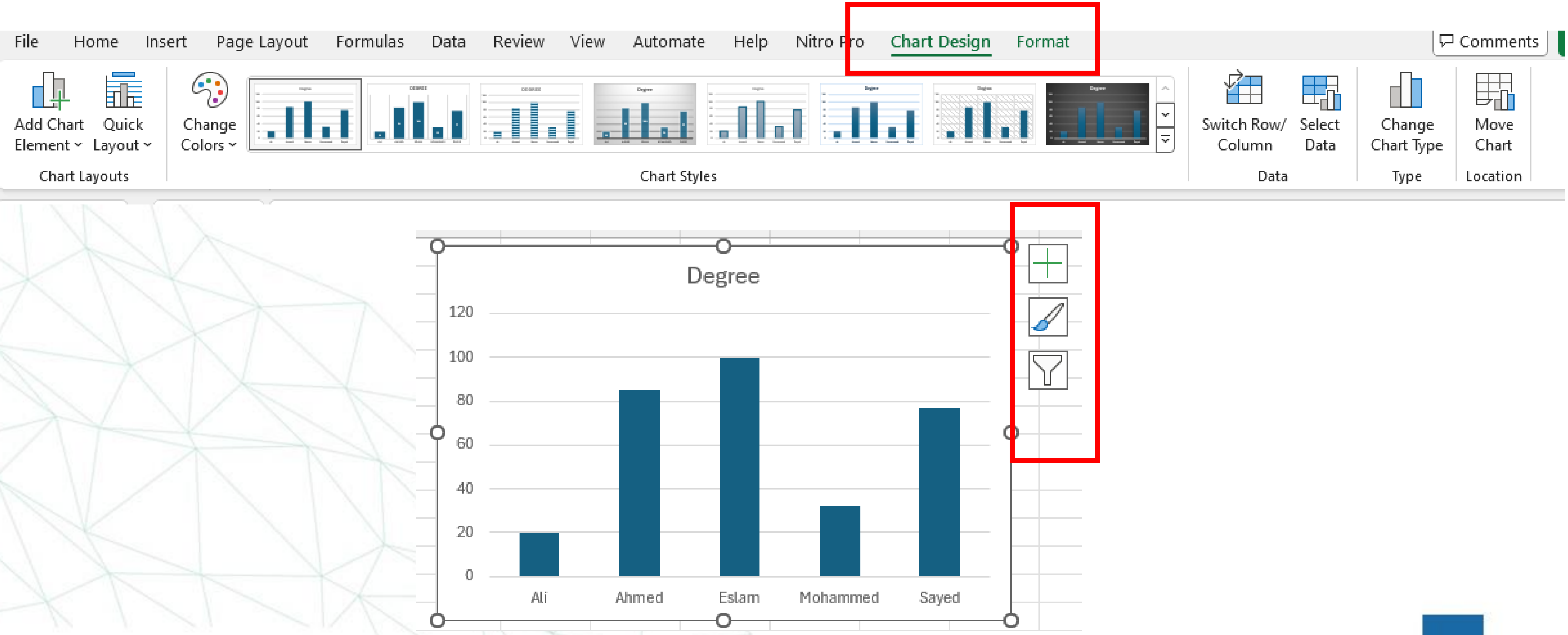
Chart Elements: Allows you to add, remove, or change chart elements like titles, labels, and gridlines.

Chart Styles: Provides various styles and color schemes to change the appearance of your chart.

Chart Filters: Lets you filter the data that is displayed in your chart.

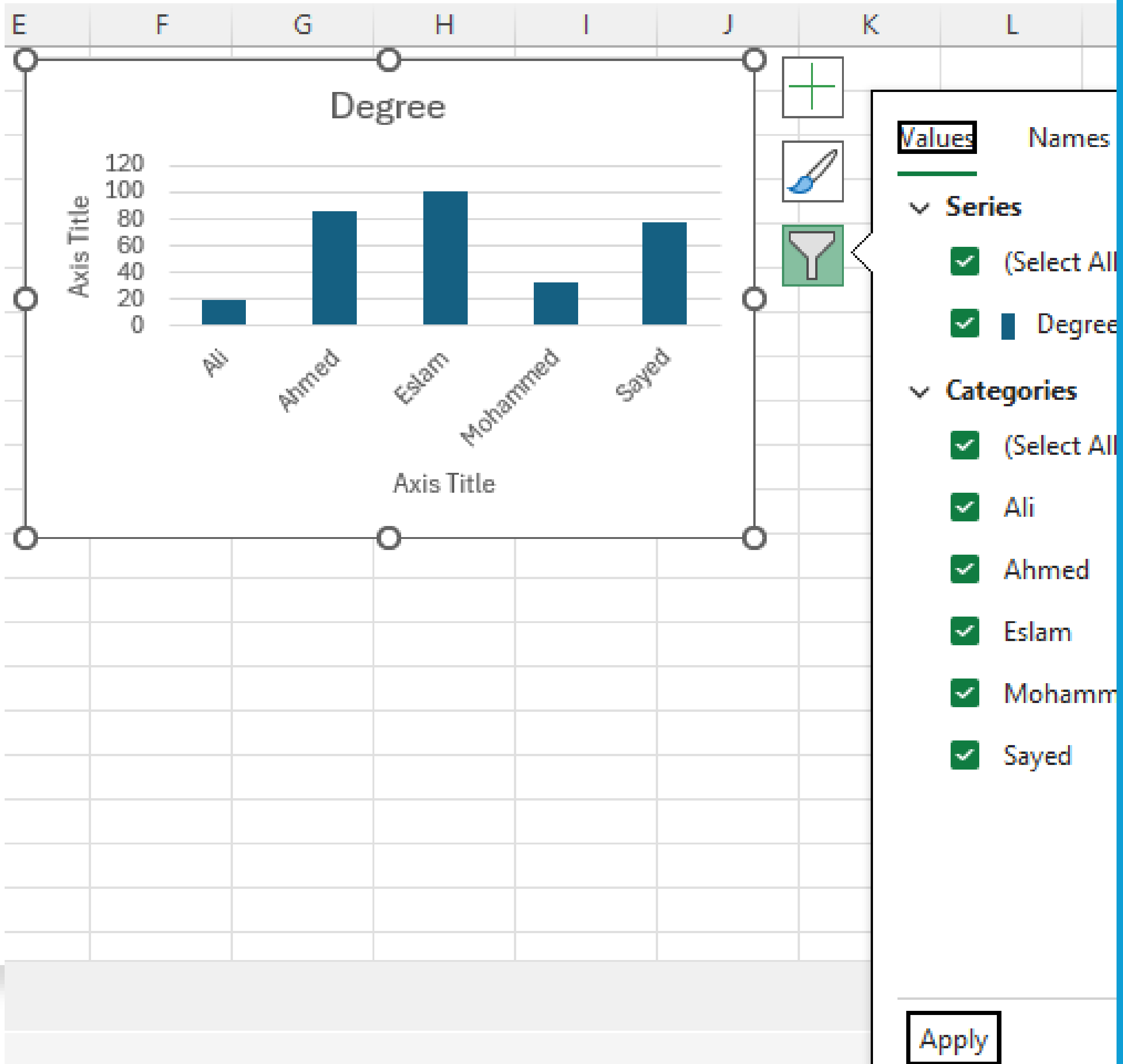
These tools help you customize the appearance and functionality of your chart to better represent your data.

Additional Options Appeared:

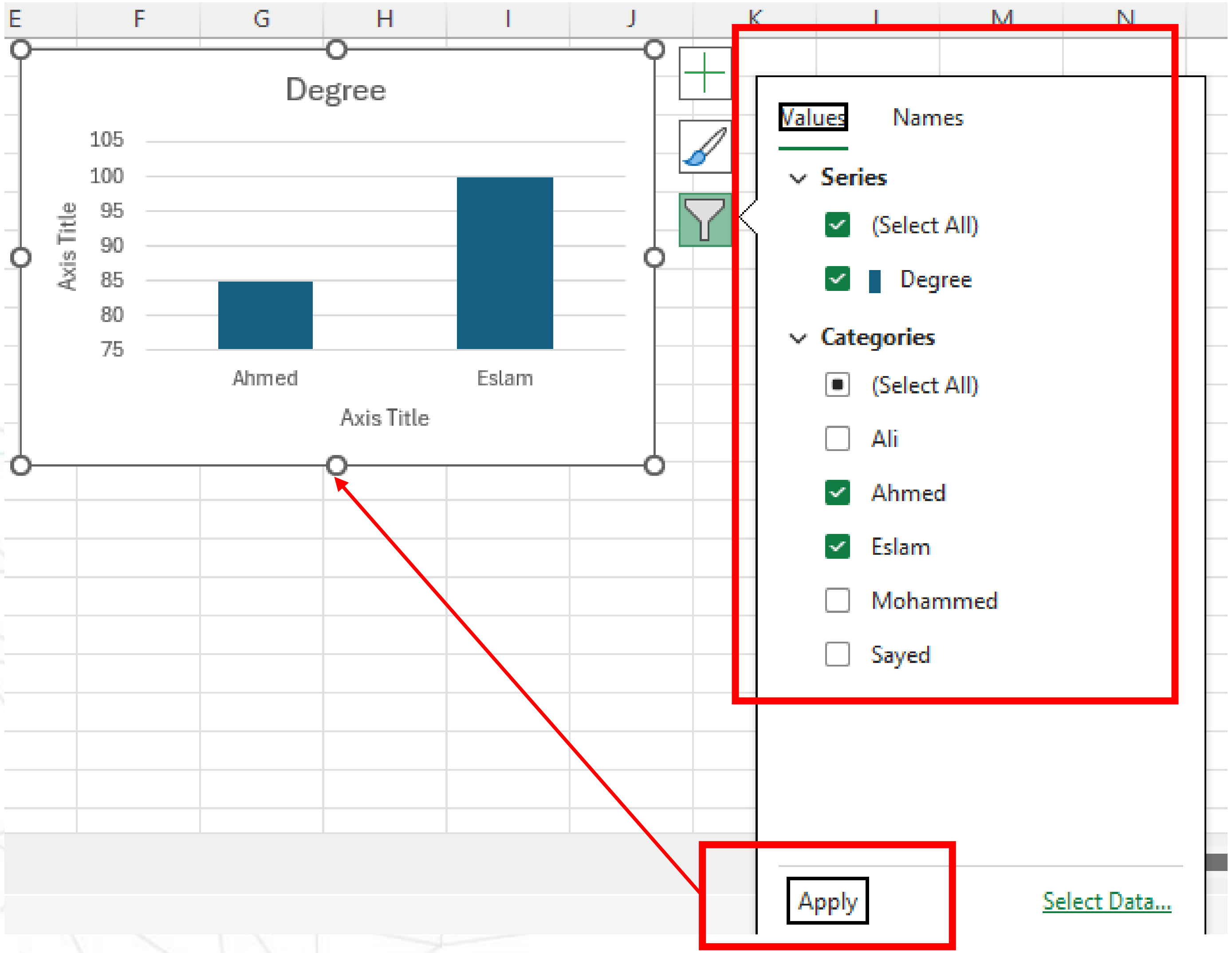


Customize Chart Filters:

- **Series**: Select or deselect data series to show or hide specific data sets in your chart. This is useful when you want to focus on certain aspects of your data without deleting any of it.
- **Categories**: Select or deselect categories to filter specific data points within the series. This allows you to narrow down the data displayed based on specific criteria.



Customize Chart Filters:

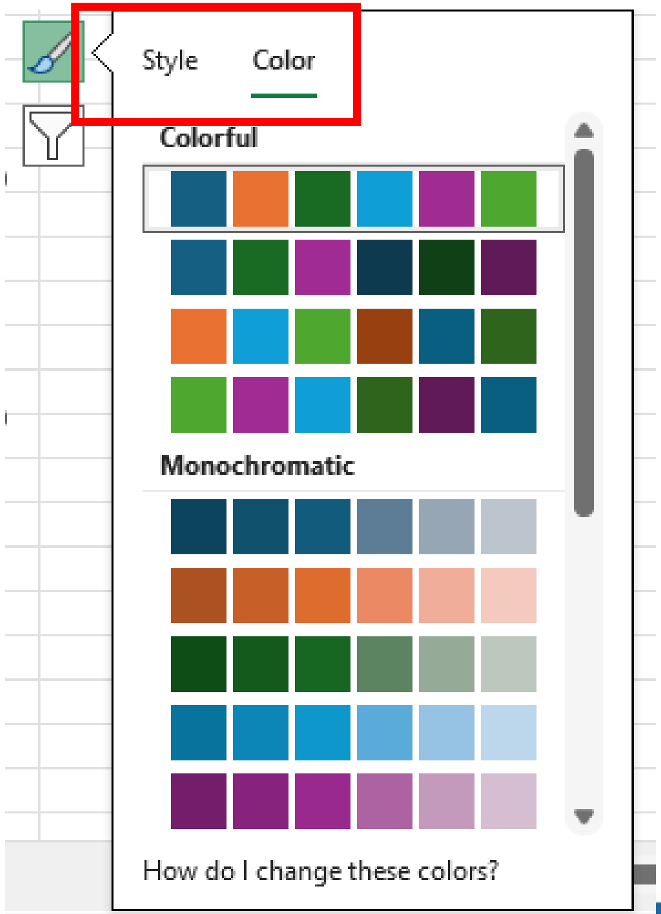


Customize Chart Styles:

Styles: Choose from a variety of predefined styles that change the overall look of your chart. These styles adjust elements such as the color, border, and background to create a polished appearance.

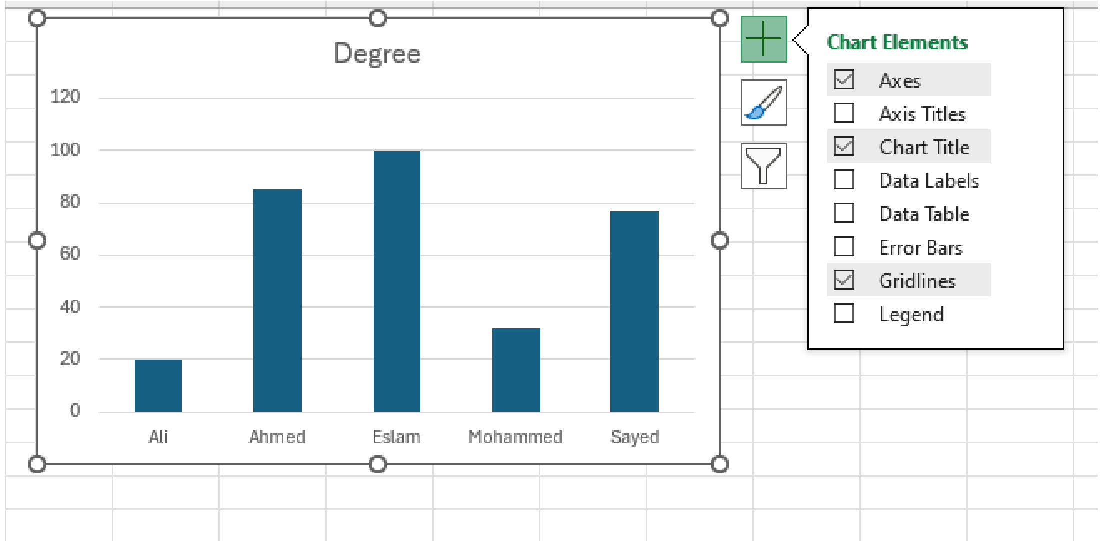
Colors: Select a color scheme that matches your presentation or report. This allows you to customize the chart's colors to better fit the theme or highlight specific data points.

Customize Chart Styles:



Customize Chart Elements:

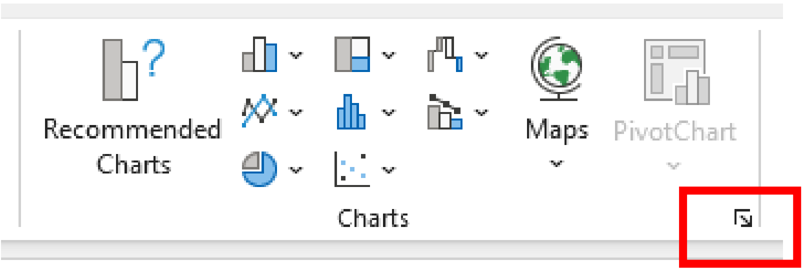
- **Axis Titles**: Add titles to the X and Y axes to clearly indicate what each axis represents.
- **Chart Title**: Add a main title to your chart to describe what the data represents.
- **Data Labels**: Display the value of each data point directly on the chart for better readability.
- **Data Table**: Include a data table below the chart to show the exact data values used.
- **Error Bars**: Add error bars to indicate the variability of the data.
- **Gridlines**: Adjust the gridlines for better visual reference points.
- **Legend**: Modify the legend to help identify different data series.
- **Trendline**: Add a trendline to show trends or patterns in the data over time.
- **Up/Down Bars**: Highlight the differences between data points.

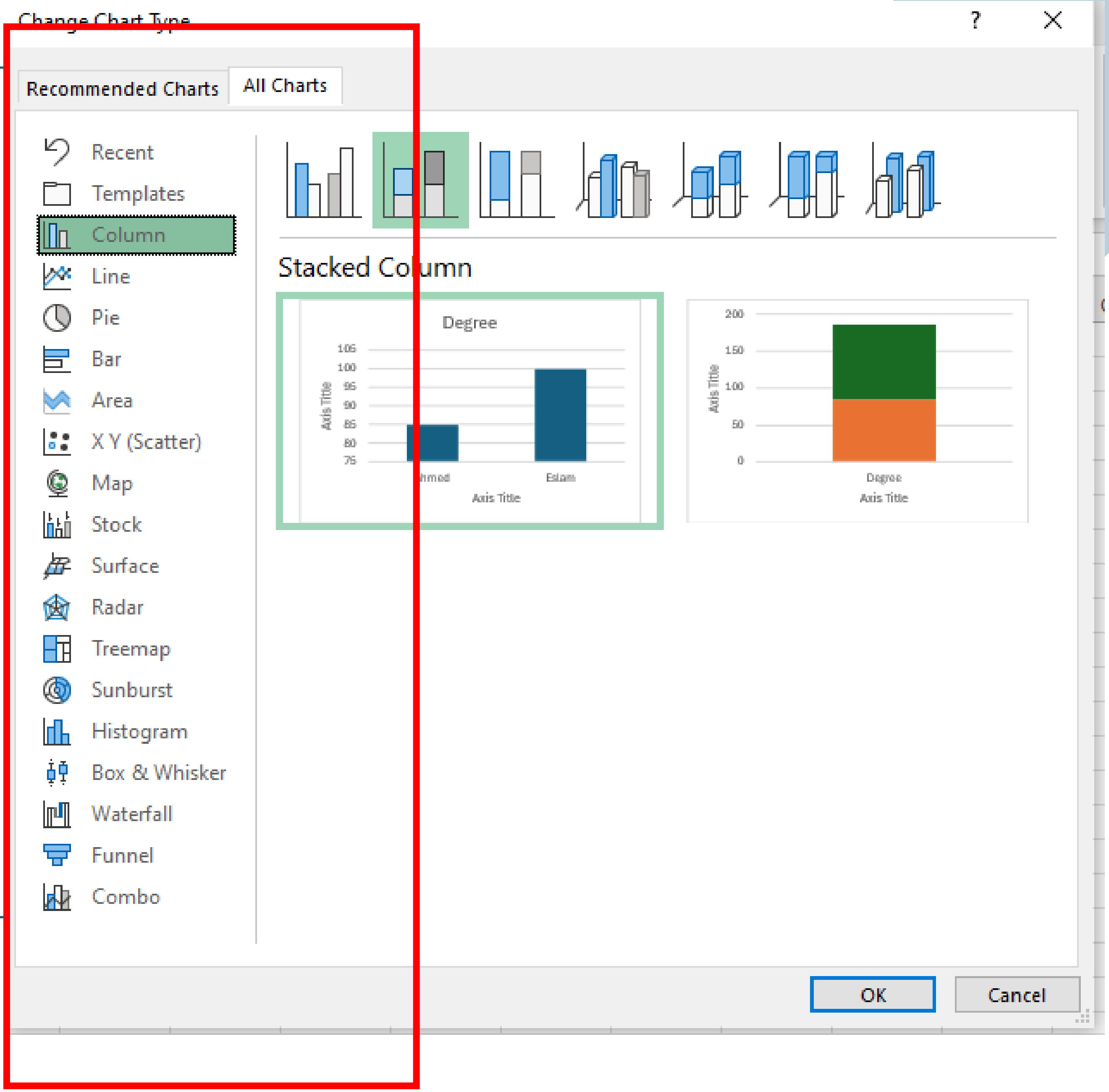


Charts Types In Excel

1. **Open Excel**:
2. **Select Your Data**:
3. **Go to the Insert Tab**:
4. **Open the Charts Menu**:
5. **Browse Chart Types**:
    a. **Recommended Charts**: Displays charts that Excel suggests based on your selected data.
    b. **All Charts**: Provides a comprehensive list of all chart types available in Excel.
6. **View All Chart Types**:
    a. **Column**: Includes clustered, stacked, and 100% stacked column charts.
    b. **Line**: Includes line, stacked line, and 100% stacked line charts.
    c. **Pie**: Includes pie, 3-D pie, and doughnut charts.
    d. **Bar**: Includes clustered bar, stacked bar, and 100% stacked bar charts.
    e. **Area**: Includes area, stacked area, and 100% stacked area charts.
    f. **Scatter**: Includes scatter plots with markers or lines.
    g. **Other**: Includes combinations, radar, and bubble charts.

In the **Charts** group, you will see several chart types displayed as icons. Click on **the small arrow** to open the full list of chart types.



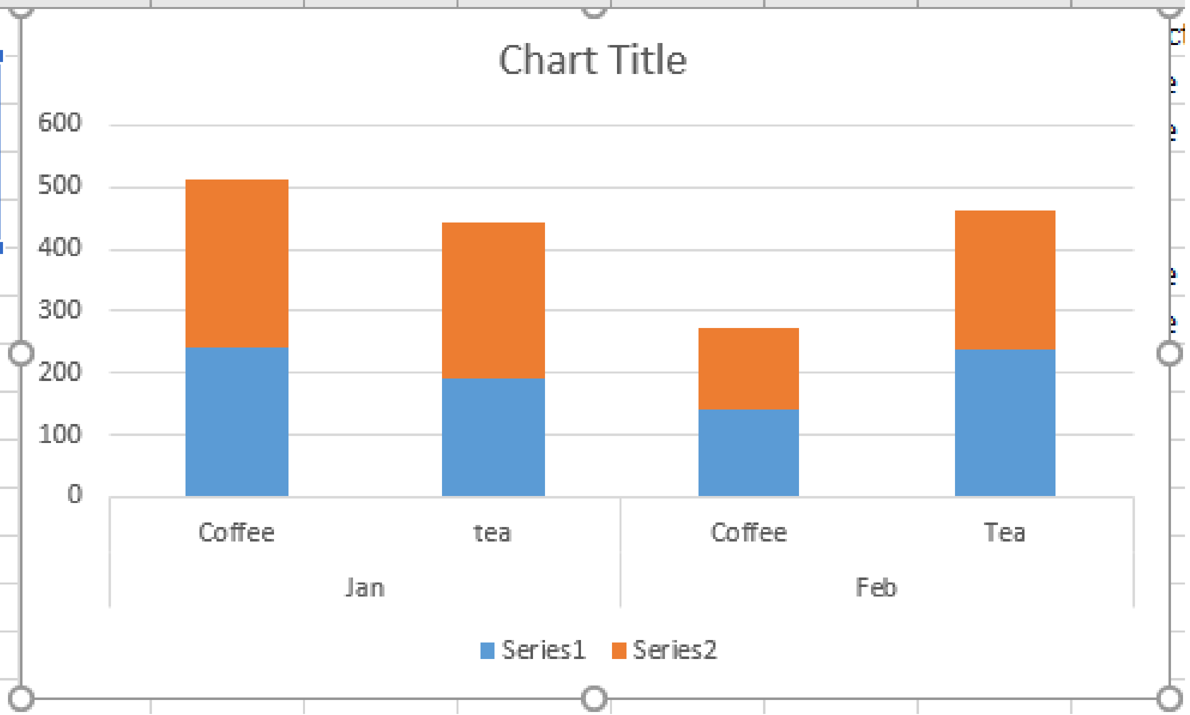


Stacked bar chart

- **Description**: A stacked bar chart displays data in horizontal bars, with each bar divided into segments representing different categories. The segments are stacked on top of each other, showing the cumulative total.
- **Use Case**: Useful for comparing the total and the contribution of individual categories over time or between different groups.
- **Example**: Showing the total sales per region, with each bar representing sales from different products.

Stacked bar chart

Month	Product	South	North
Jan	Coffee	241	273
	tea	191	253
Feb	Coffee	139	133
	Tea	238	224



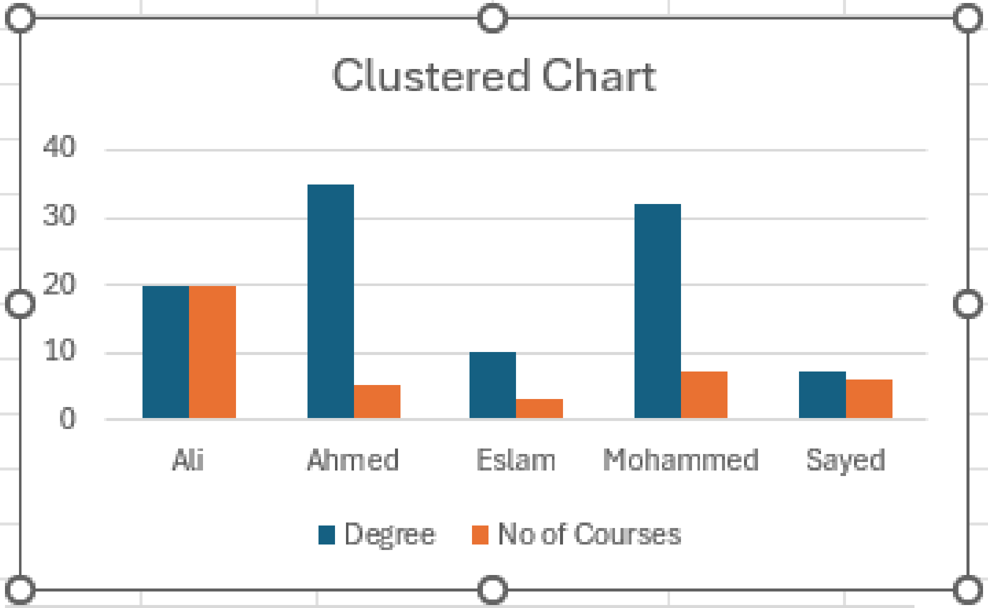
Cluster bar chart

- **Description**: A cluster bar chart displays data in horizontal bars grouped together by category. Each cluster represents a category with multiple bars within it, showing the values for different sub-categories.
- **Use Case**: Ideal for comparing multiple items across different categories, where each category is represented by a cluster of bars.
- **Example**: Comparing sales figures of different products across several quarters.

Cluster bar chart

Chart 2

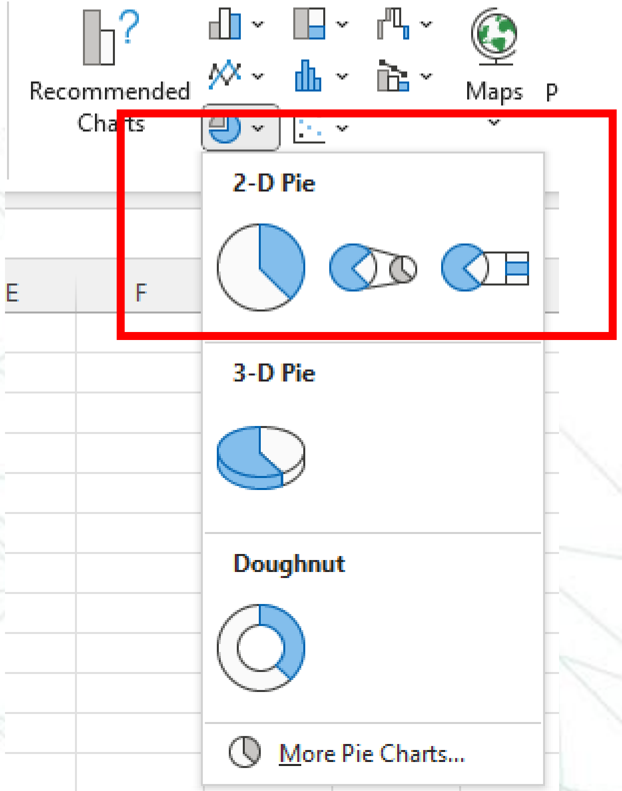
	A	B	C
1	Names	Degree	No of Courses
2	Ali	20	20
3	Ahmed	35	5
4	Eslam	10	3
5	Mohammed	32	7
6	Sayed	7	6

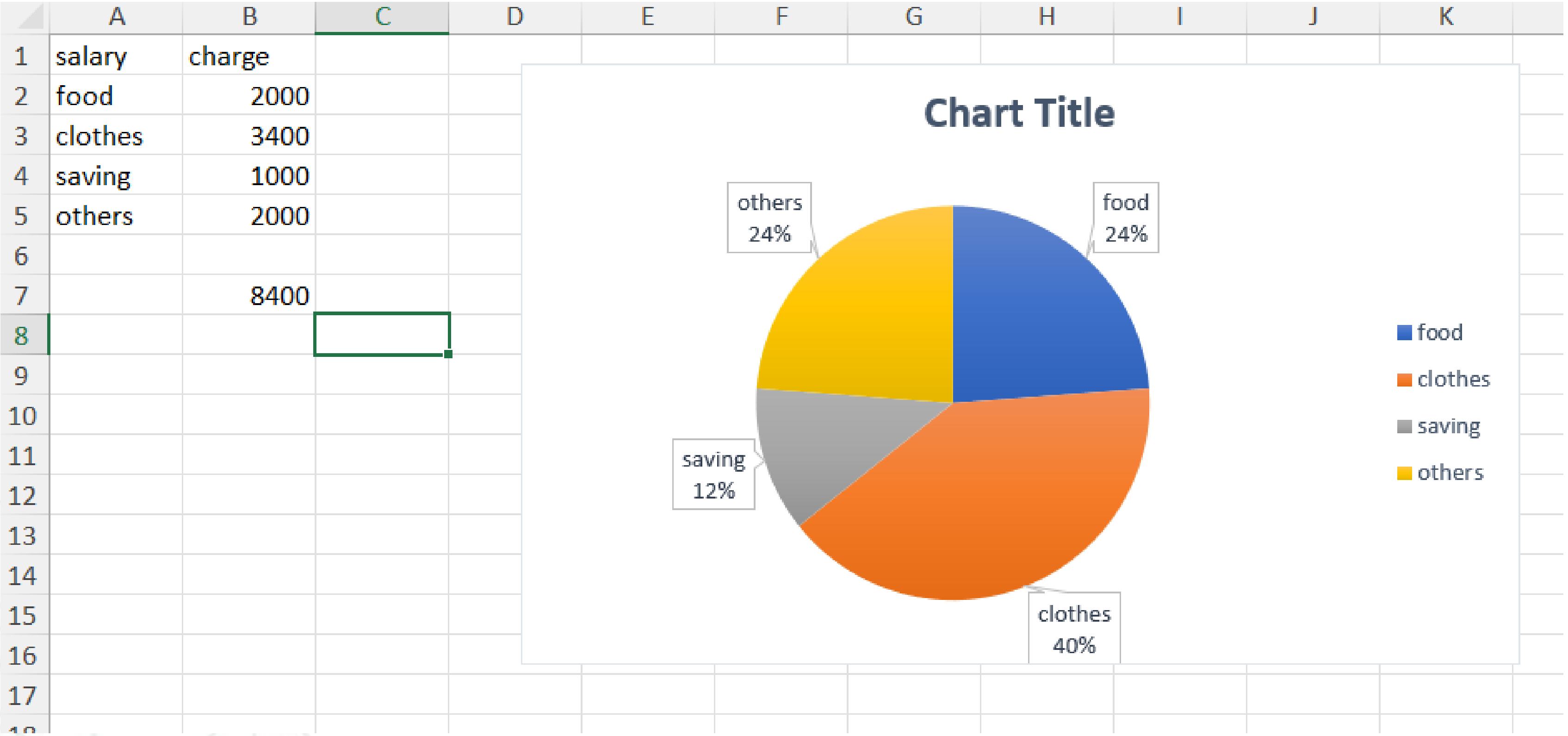


Pie chart

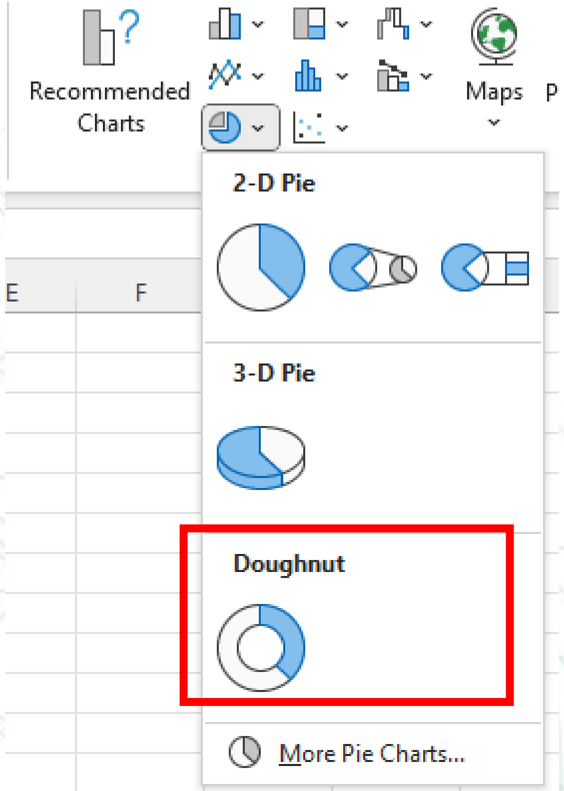
- **Description**: A pie chart represents data as slices of a circle, where each slice corresponds to a category’s proportion of the total. The size of each slice is proportional to the category’s percentage of the total.
- **Use Case**: Best for showing the relative size of parts to a whole, particularly when you want to illustrate proportions or percentages.
- **Example**: Displaying the market share of different companies within an industry.

Pie chart

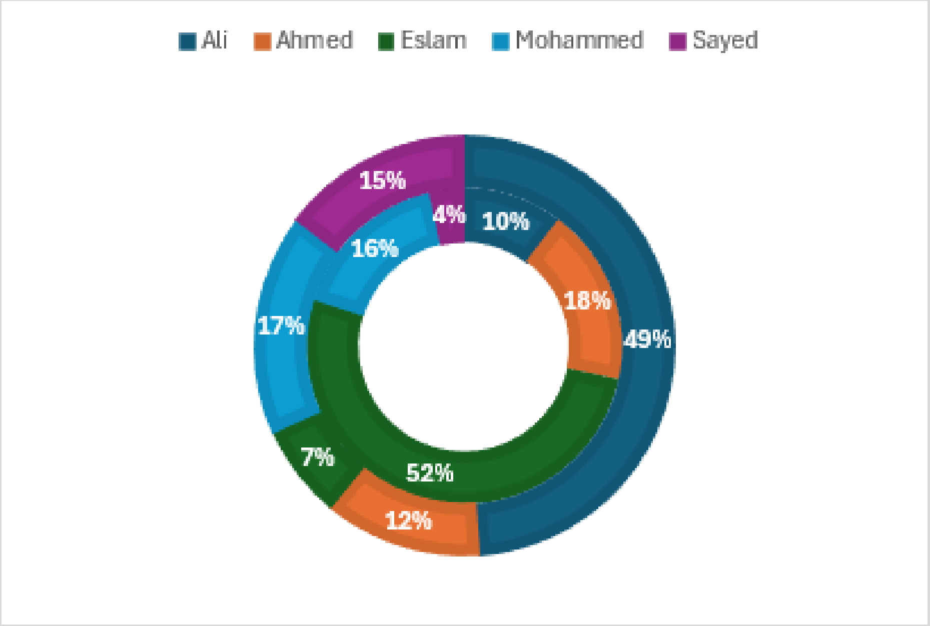

Recommended Charts
Maps
2-D Pie
3-D Pie
Doughnut
More Pie Charts...


salary charge
food 2000
clothes 3400
saving 1000
others 2000
8400
Chart Title
others 24%
food 24%
saving 12%
clothes 40%
food
clothes
saving
others

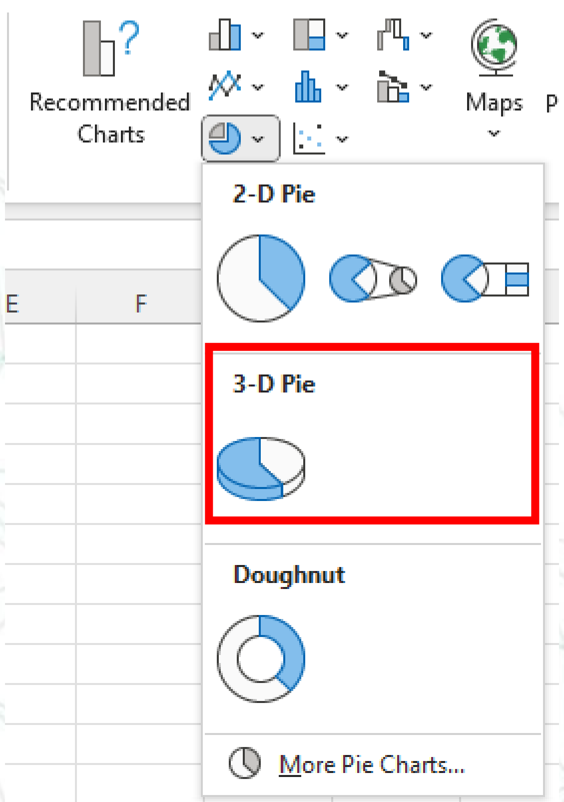
Pie chart



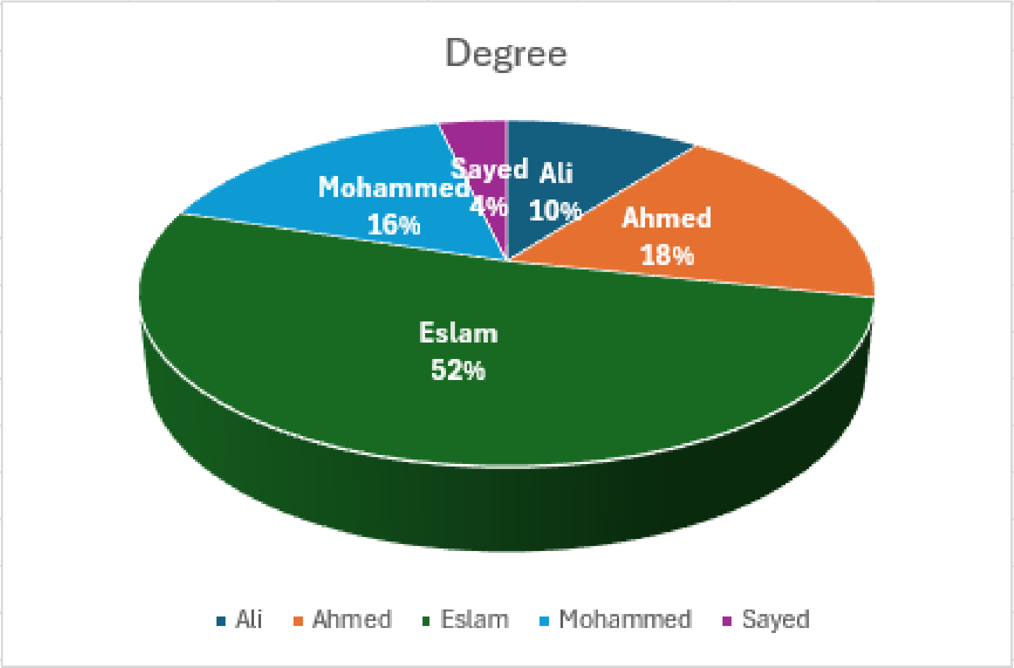
Names	Degree	No of Courses
Ali	20	20
Ahmed	35	5
Eslam	100	3
Mohammed	32	7
Sayed	7	6



Pie chart



	A	B	C
1	Names	Degree	No of Courses
2	Ali	20	20
3	Ahmed	35	5
4	Eslam	100	3
5	Mohammed	32	7
6	Sayed	7	6

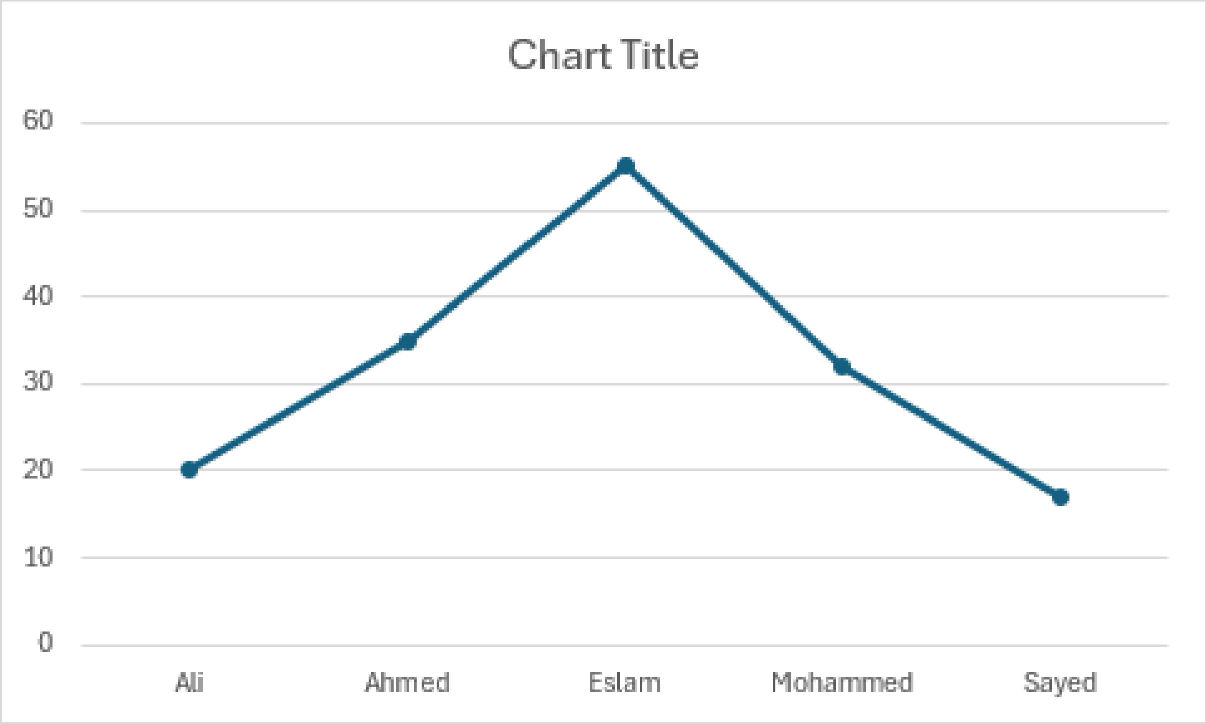


Line chart

- **Description**: A line chart plots data points along a continuous line, showing trends over time or categories. It's useful for visualizing changes and trends.
- **Use Case**: Ideal for tracking performance, trends, or changes over a period.
- **Example**: Tracking monthly revenue growth over a year.

Line chart

Names	Degree	No of Courses
Ali	20	1
Ahmed	35	15
Eslam	55	16
Mohammed	32	7
Sayed	17	6

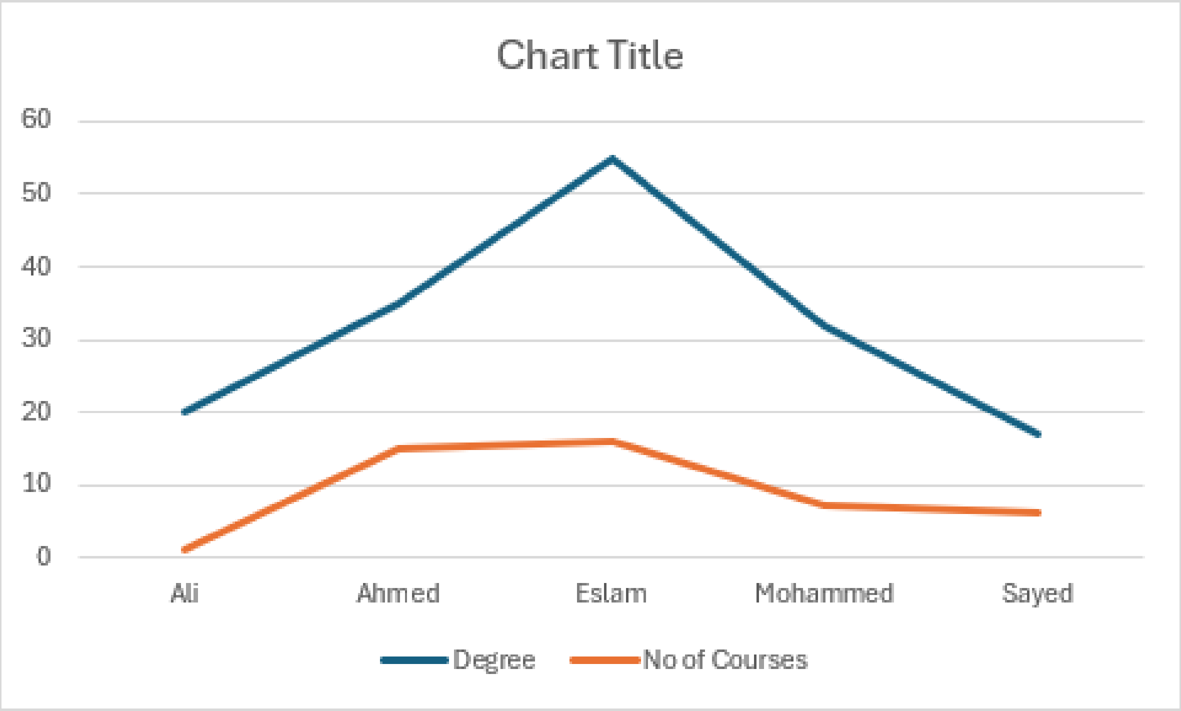


Multiple line chart:

- **Description**: A multiple line chart is a line chart with more than one line, each representing a different data series. It allows you to compare multiple trends on the same graph.
- **Use Case**: Useful for comparing different sets of data over time or categories.
- **Example**: Comparing the sales trends of different products over several months.

Multiple line chart:

	A	B	C
1	Names	Degree	No of Courses
2	Ali	20	1
3	Ahmed	35	15
4	Eslam	55	16
5	Mohammed	32	7
6	Sayed	17	6

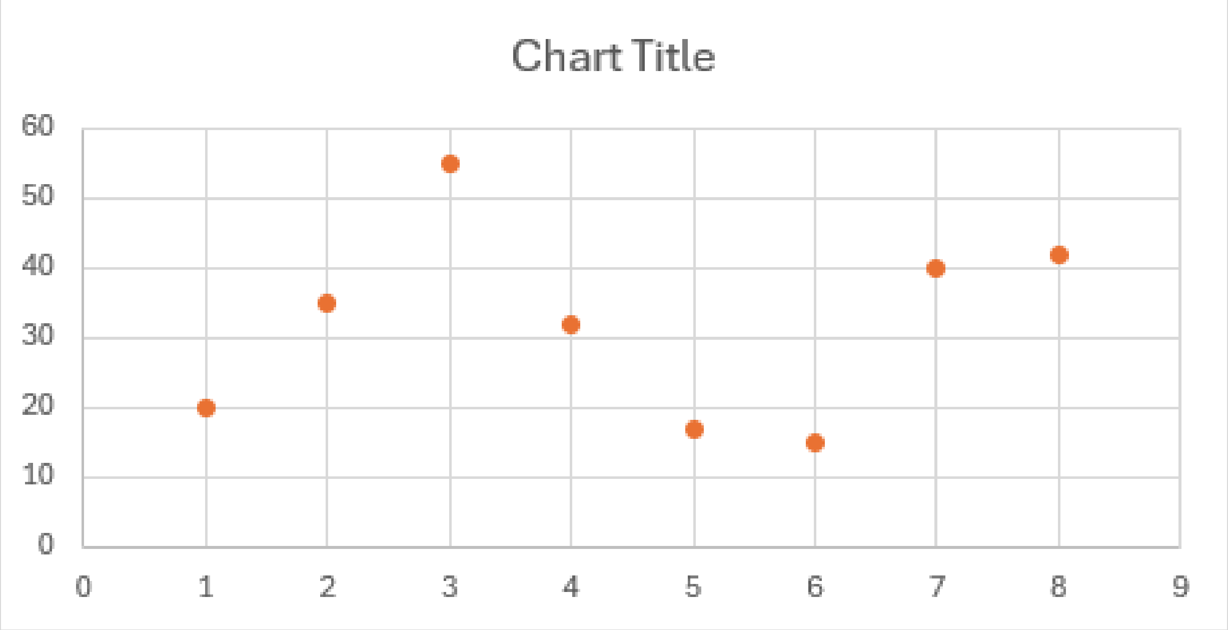


Scatter plot

- **Description**: A scatter plot displays data points as individual markers on a Cartesian plane, with one variable plotted on the X-axis and another on the Y-axis. It helps in identifying relationships or correlations between two variables.
- **Use Case**: Ideal for examining relationships between two numerical variables and identifying trends, patterns, or outliers.
- **Example**: Analyzing the correlation between advertising spend and sales revenue.

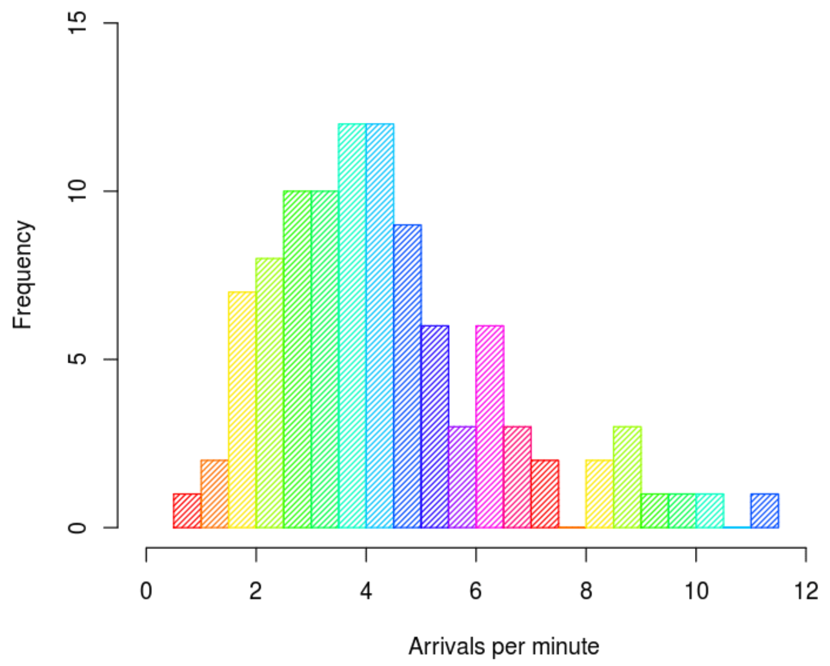
Scatter plot

Names	Degree
Ali	20
Ahmed	35
Eslam	55
Mohammed	32
Sayed	17
aya	15
yassin	40
omar	42



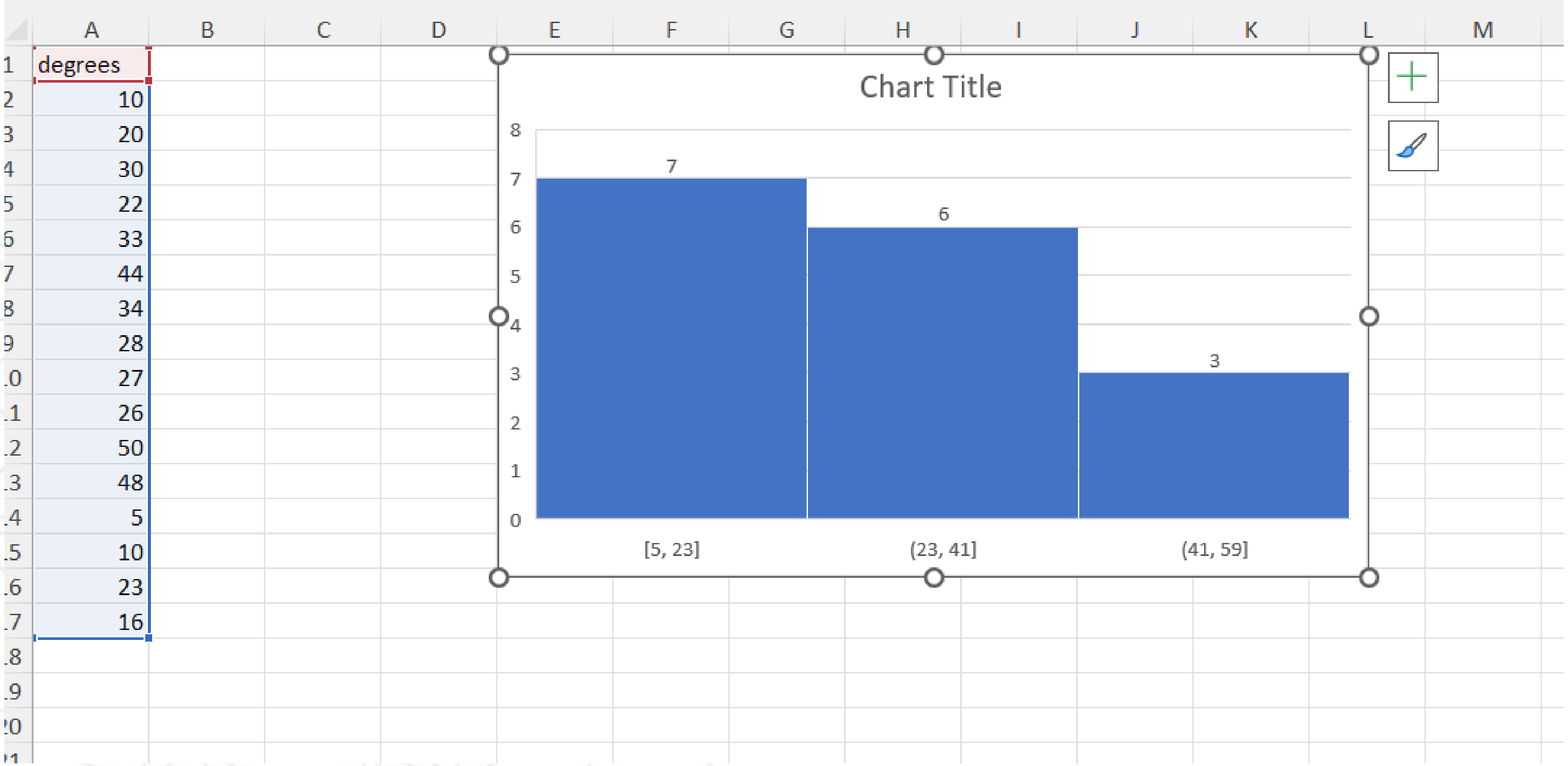
Histogram

- **Purpose**: A Histogram is used to display the distribution of a dataset by showing the frequency of data points within certain ranges or "bins."
- **When to Use**:
  - When you want to understand the distribution of continuous data (e.g., test scores, age groups).
  - When analyzing the shape of data distribution, such as identifying skewness or identifying outliers.
- **How It Works**:
  - Data is divided into bins (ranges), and the height of each bar represents the number of data points within that bin.
- **Example**: Visualizing the distribution of sales data over time to identify patterns like most frequent sales ranges.



Histogram

In Excel, you can create a Histogram by selecting your data, then navigating to the “Insert” tab, choosing "Histogram" from the "Charts" group.

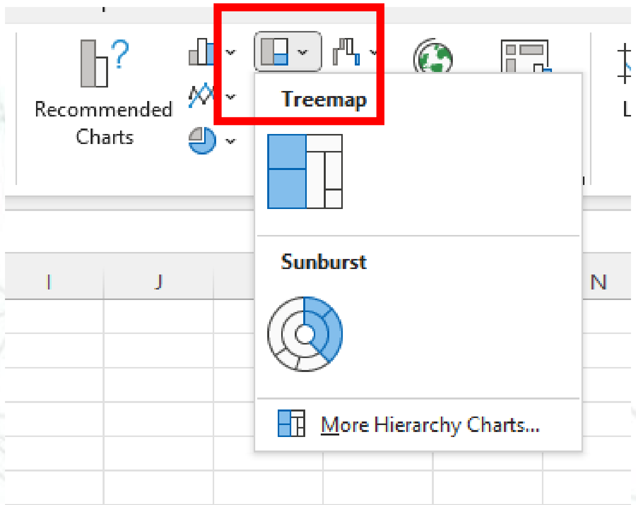


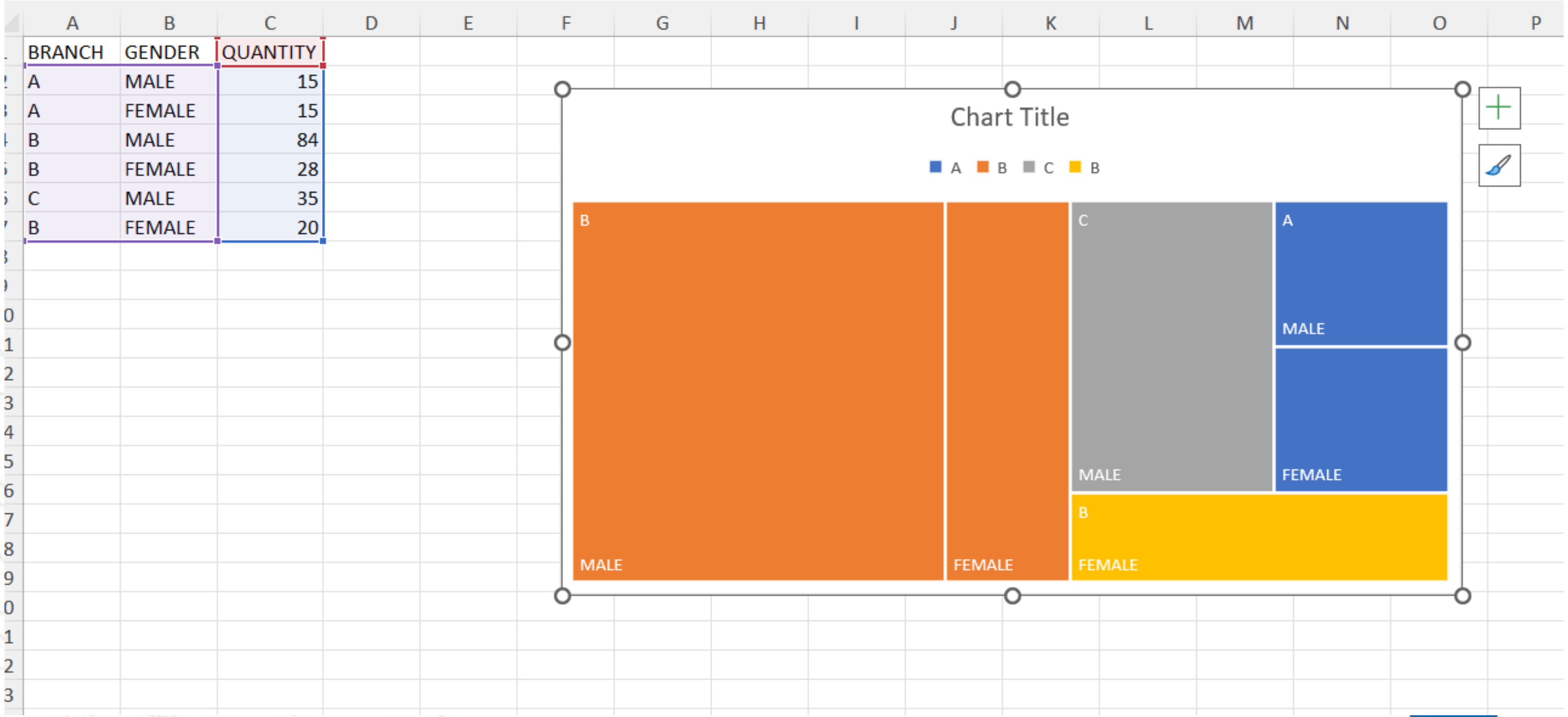
Tree map

- **Purpose:** A Tree Map visualizes hierarchical data as a set of nested rectangles, where the size of each rectangle is proportional to the value it represents.
- **When to Use:**
  - When you want to compare the proportions of different categories within a hierarchy.
  - When visualizing data with multiple levels of categories (e.g., market share by company, then by product).
- **How It Works:**
  - The rectangles represent different categories, with larger rectangles indicating higher values. The hierarchy is shown by nesting smaller rectangles within larger ones.
- **Example:** Visualizing the revenue contribution of various departments within a company, where each department is broken down into its respective products or services.

Tree map

- To create a Tree Map in Excel, select your hierarchical data and choose "Tree Map" from the “Insert” tab under the “Hierarchy Charts" group.





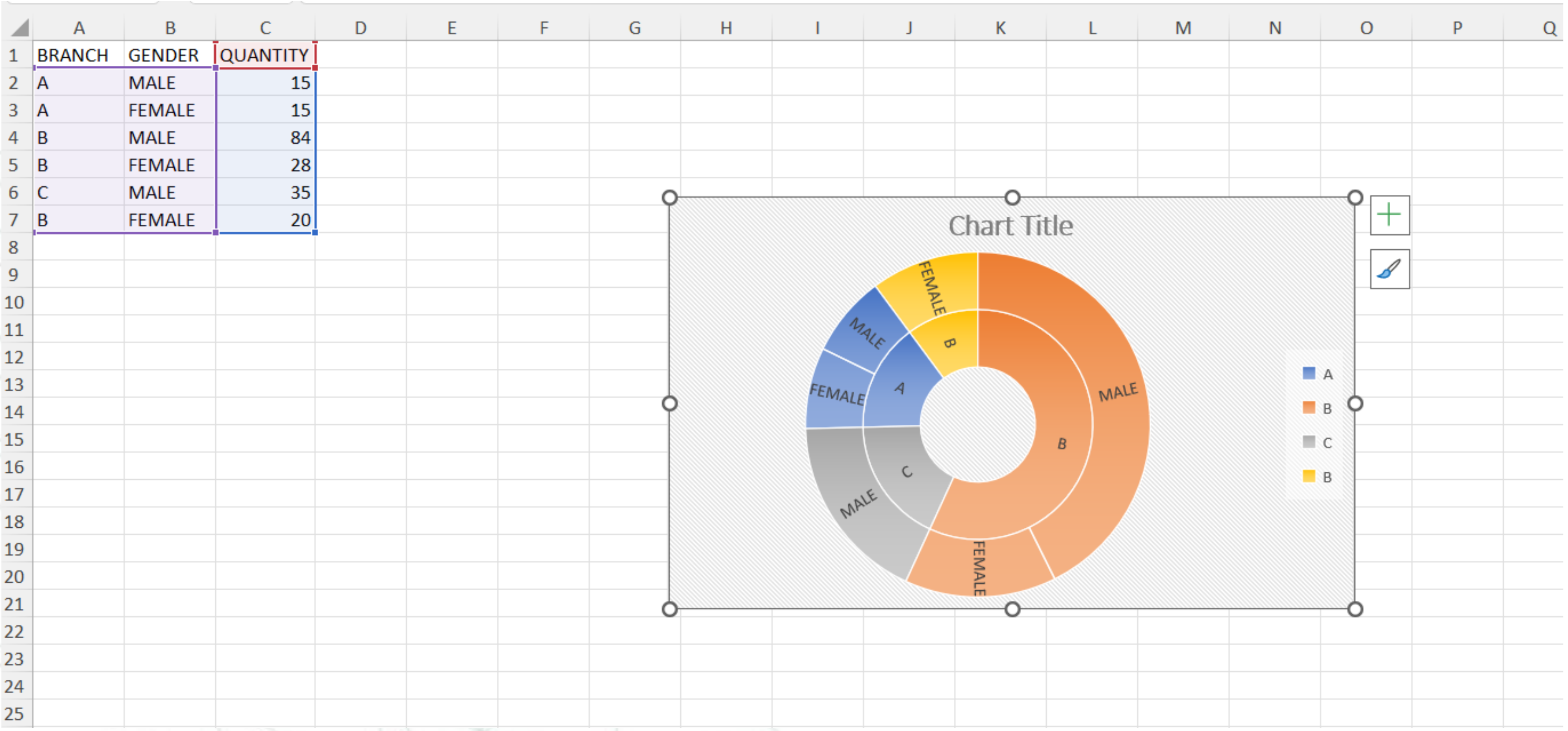
Sun Burst

- **Purpose**: A Sunburst Chart is used to visualize hierarchical data in a circular format. It is like a Tree Map but with a radial layout, where each level of the hierarchy is represented by a ring or segment around a central point.
- **When to Use**:
  - When you want to show how each level of a hierarchy contributes to the whole.
  - When highlighting both the structure of the hierarchy and the proportion of each category.
- **How It Works**:
  - The innermost circle represents the top-level category, with subsequent rings representing deeper levels of the hierarchy. The size of each segment corresponds to its proportion within the hierarchy.
  - The chart allows for easy identification of relationships and contributions across different levels.

Example: Visualizing a company's organizational structure, where the inner circle represents the overall company, and each subsequent ring represents departments, teams, and individual employees. Another example could be breaking down sales data by region, country, and product category

Sun burst

To create a Sunburst Chart in Excel, select your hierarchical data, go to the “Insert” tab, and choose "Sunburst Chart" from the "Hierarchy" charts group. Ensure your data is structured in a way that represents the hierarchy you want to display.



BRANCH	GENDER	QUANTITY
A	MALE	15
A	FEMALE	15
B	MALE	84
B	FEMALE	28
C	MALE	35
B	FEMALE	20

Comparison with Other Hierarchical Charts (Tree Map and Sun burst)

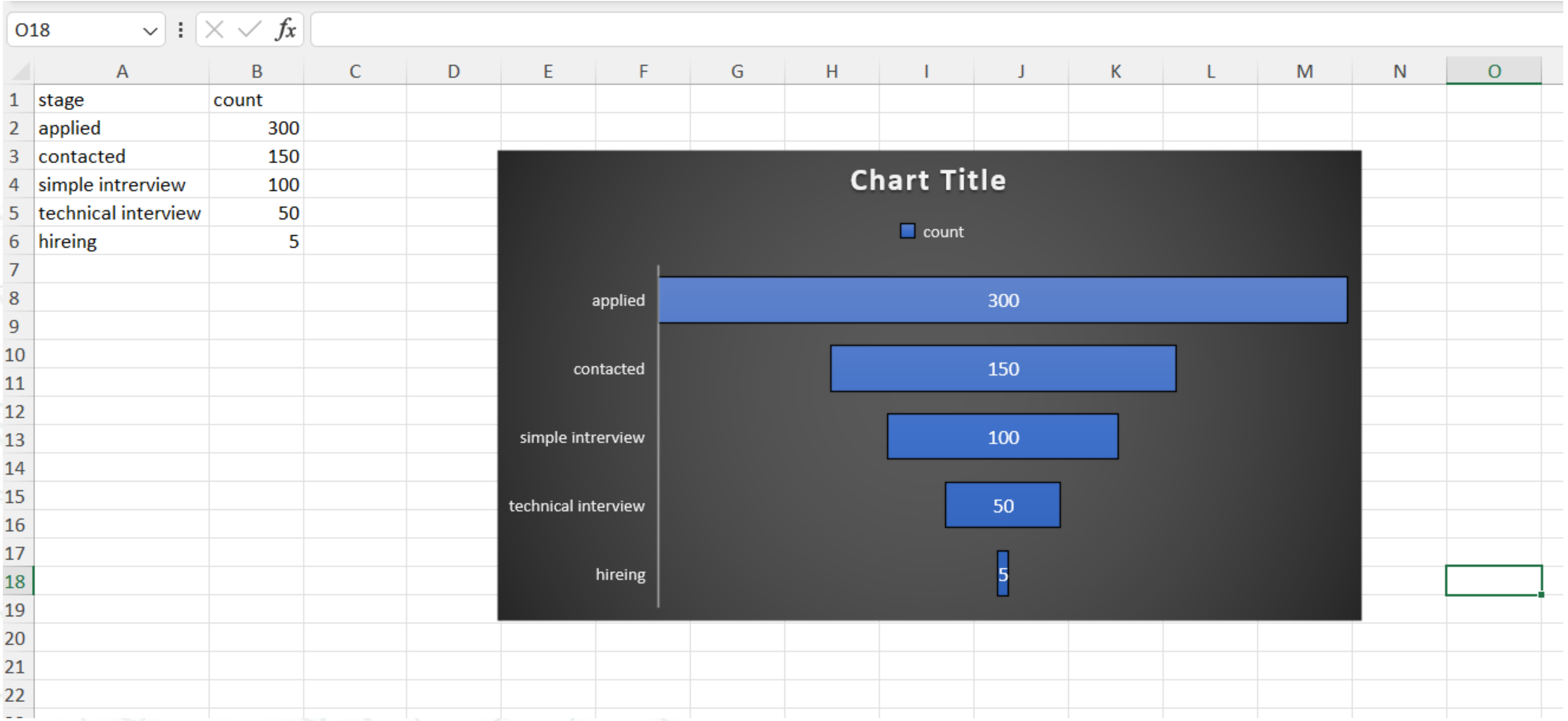
- **Tree Map vs. Sunburst Chart**:
  - Both charts are used for hierarchical data, but the Sunburst Chart offers a more visual and intuitive representation for certain datasets, especially when you want to emphasize the structure of the hierarchy.
  - Tree Maps are more suited for comparing the size of categories within the hierarchy, while Sunburst Charts are better for displaying the hierarchical structure itself.

Funnel chart

- **Purpose:** Funnel charts are used to represent data that flows through stages in a process, with each stage represented by a segment of the funnel.
- **When to Use:**
  - When you want to visualize the reduction of data as it passes through stages in a linear process (e.g., sales pipeline, lead conversion).
  - When identifying where the largest drop-offs occur in a process.
- **How It Works:**
  - The chart narrows down from the top to the bottom, with each segment representing a stage. The size of the segment reflects the proportion of the data remaining at that stage.
- **Example:** Visualizing the stages of a sales funnel, from initial contact to final purchase, showing where potential customers drop out.

Funnel chart

Funnel charts are available under the “Insert” tab. Select your data, and then choose "Funnel" from the "Charts" group.



stage	count
applied	300
contacted	150
simple intrerview	100
technical interview	50
hireing	5

Waterfall chart

- **Purpose:** A Waterfall Chart is used to show the cumulative effect of sequential positive and negative values on a starting point. It is particularly useful for understanding how individual components contribute to a total, often used in financial analysis.
- **When to Use:**
  - When you need to visualize the step-by-step contribution of various elements to a final value (e.g., starting from a net income, showing how revenues, costs, and other factors lead to the final profit).
  - When analyzing the impact of different components on a single measure, such as analyzing changes in profit over time or the effect of various cost factors on a budget.
  - When you want to highlight the contributions of increases and decreases to the overall change.

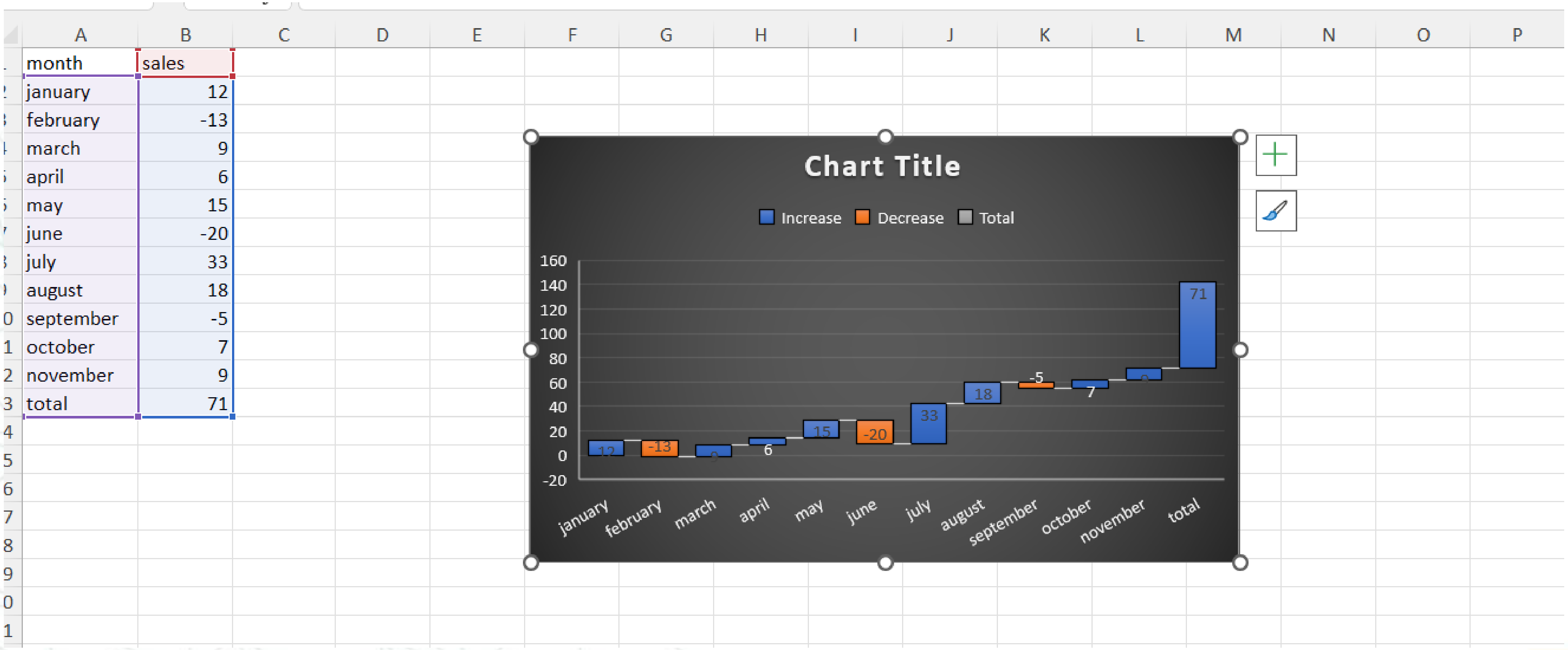
Waterfall chart

- **How It Works:**
  - The chart starts with an initial value, and subsequent bars represent the additions (increases) or subtractions (decreases) that lead to the final total.
  - Positive values are typically shown in one color, rising from the previous bar, while negative values are shown in another color, dropping from the previous bar. The final bar usually represents the net total after all additions and subtractions.
- **Example:** Visualizing a company's financial performance over a quarter, starting with the previous quarter's profit, then adding revenues, subtracting various costs (like COGS, operating expenses), and ending with the net profit for the current quarter.

Waterfall chart

Select your data, go to the “Insert” tab, and choose "Waterfall Chart" from the "Charts" group.

Excel will automatically create a Waterfall Chart, distinguishing positive and negative contributions with different colors.



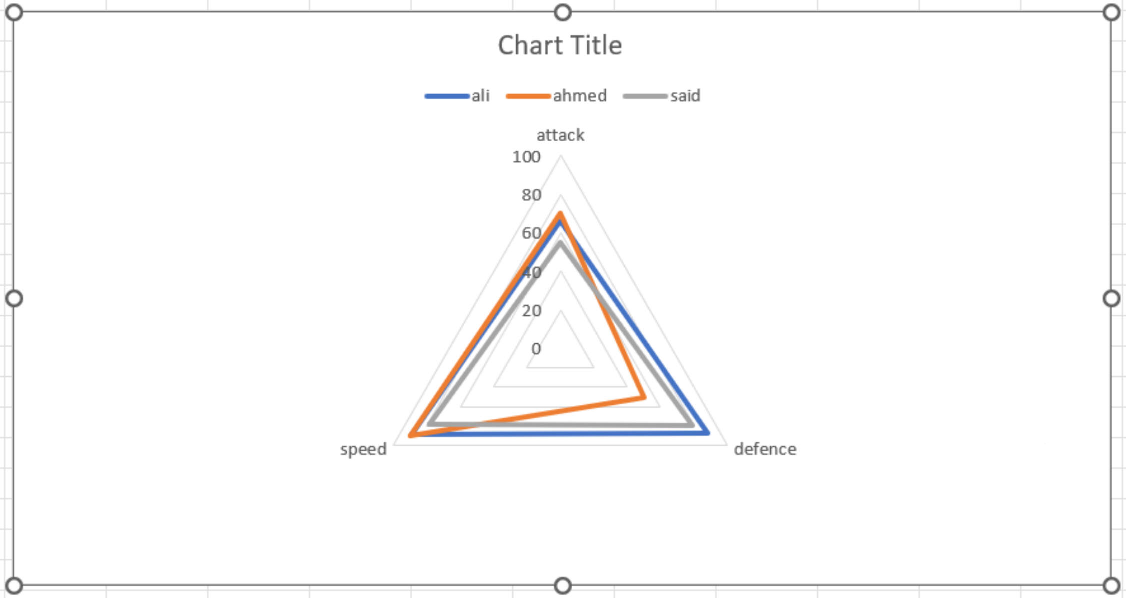
Radar chart

- **Purpose:** A Radar Chart (also known as a spider or web chart) is used to compare multiple quantitative variables. Each variable has its own axis, which radiates out from the center.
- **When to Use:**
  - When comparing the relative performance of different entities on various metrics (e.g., performance evaluations, feature comparisons).
  - When identifying strengths and weaknesses across different categories.
- **How It Works:**
  - Each data point is plotted on its corresponding axis, and the points are connected to form a polygon. The size and shape of the polygon provide insights into the data.
- **Example:** Comparing the performance metrics of different departments in an organization, such as customer satisfaction, cost efficiency, and innovation.

Radar chart

To create a Radar Chart, select your data, navigate to the “Insert” tab, and choose "Radar Chart" from the "Charts" group.

name	attack	defence	speed
ali	66	88	89
ahmed	70	50	90
said	55	79	78



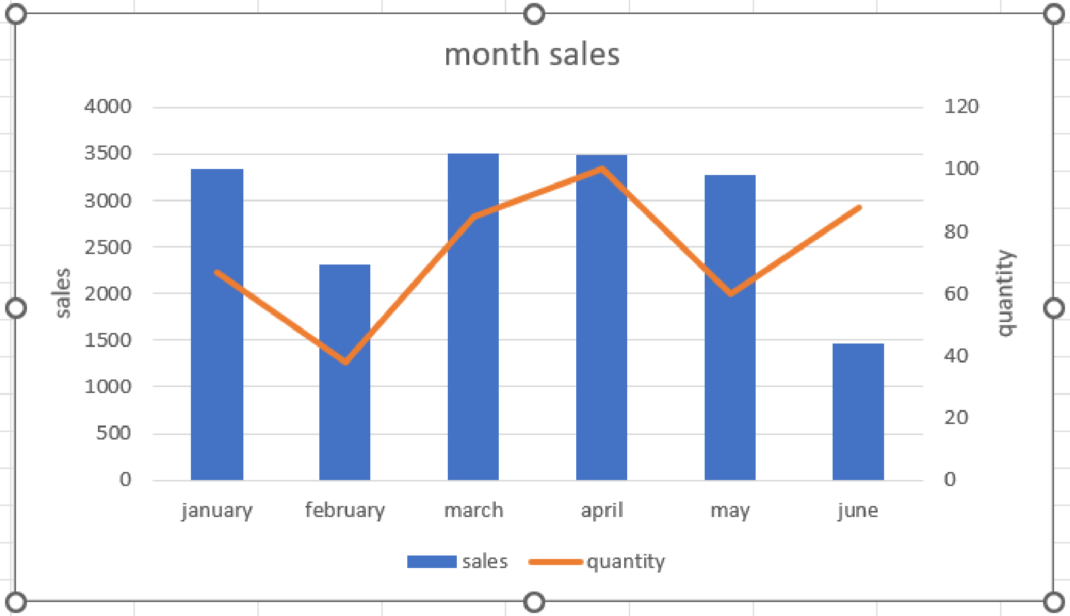
Combo chart

- **Purpose:** A Combo Chart combines two or more different types of charts (e.g., line and bar) to compare different datasets or highlight different aspects of a dataset.
- **When to Use:**
  - When you want to compare different types of data in one chart (e.g., revenue as bars and profit margin as a line).
  - When displaying data that has different scales or when emphasizing the relationship between datasets.
- **How It Works:**
  - One dataset is represented by one type of chart (e.g., bars for sales), and another dataset by a different type (e.g., a line for profit margin) on the same axis.
- **Example:** Comparing monthly revenue (as bars) against profit margin (as a line) to see how changes in revenue impact profitability.

Combo chart

In Excel, create a Combo Chart by selecting your data, choosing "Insert," then "Combo Chart," and customize by selecting which series should be bars, lines, etc.

month	sales	quantity
january	3334	67
february	2312	38
march	3500	85
april	3492	100
may	3268	60
june	1471	88



Exercise 1: Comparing Sales Data Across Multiple Regions

- **Task :** Use a bar or column chart to compare quarterly sales data across different regions.
- **Challenge :** Highlight the top-performing region in a different color.

Exercise 2: Analyzing Customer Trends Over Time

- **Task:** Create a line chart to visualize how customer preferences have changed over the past year.
- **Challenge:** Add a trendline to show overall trends and make predictions.

Exercise 3: Displaying Distribution of Customer Age Groups

- **Task:** Use a histogram to display the distribution of customer ages.
- **Challenge:** Adjust bin sizes for optimal clarity and interpretation.